



**QUEEN'S
UNIVERSITY
BELFAST**

Evaluation and optimisation of pre-trained CNN models for asphalt pavement crack detection and classification

Matarneh, S., Elghaish, F., Pour Rahimian, F., Abdellatef, E., & Abrishami, S. (2024). Evaluation and optimisation of pre-trained CNN models for asphalt pavement crack detection and classification. *Automation in Construction*, 160, Article 105297. <https://doi.org/10.1016/j.autcon.2024.105297>

Published in:
Automation in Construction

Document Version:
Publisher's PDF, also known as Version of record

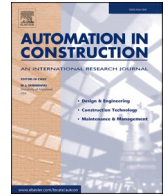
Queen's University Belfast - Research Portal:
[Link to publication record in Queen's University Belfast Research Portal](#)

Publisher rights
Copyright 2024 the authors.
This is an open access article published under a Creative Commons Attribution License (<https://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution and reproduction in any medium, provided the author and source are cited.

General rights
Copyright for the publications made accessible via the Queen's University Belfast Research Portal is retained by the author(s) and / or other copyright owners and it is a condition of accessing these publications that users recognise and abide by the legal requirements associated with these rights.

Take down policy
The Research Portal is Queen's institutional repository that provides access to Queen's research output. Every effort has been made to ensure that content in the Research Portal does not infringe any person's rights, or applicable UK laws. If you discover content in the Research Portal that you believe breaches copyright or violates any law, please contact openaccess@qub.ac.uk.

Open Access
This research has been made openly available by Queen's academics and its Open Research team. We would love to hear how access to this research benefits you. – Share your feedback with us: <http://go.qub.ac.uk/oa-feedback>



Evaluation and optimisation of pre-trained CNN models for asphalt pavement crack detection and classification

Sandra Matarnah^a, Faris Elghaish^{b,*}, Farzad Pour Rahimian^{c,*}, Essam Abdellatif^d, Sepehr Abrishami^e

^a Faculty of Engineering, Al Ahliyya Amman University, Jordan

^b School of Natural and Built Environment, Queen's University Belfast, Belfast, UK

^c School of Computing, Engineering & Digital Technologies, Teesside University, Middlesbrough, UK

^d Department of Electrical Engineering, Faculty of Engineering, Sinai University, El-Arish 45511, Egypt

^e School of Civil Engineering and Surveying, University of Portsmouth, Portsmouth, UK

ARTICLE INFO

Keywords:

Asphalt pavement
Cracks classification
CNN
OpDenseNet201
Optimisation
Deep learning
Pre-trained models

ABSTRACT

This study explored the performance of ten pre-trained CNN architectures in detecting and classifying asphalt pavement cracks from images. A comparison of eight optimisation techniques led to developing an optimised pre-trained CNN model tailored for crack classification, with DenseNet201 emerging as the most effective, closely followed by ShuffleNet and ResNet101. Conversely, VGG16 exhibited notably lower accuracy among the models evaluated. Through the application of diverse feature selection techniques as optimisers, DenseNet201 consistently outperformed others, followed by DarkNet19 and Xception. Despite employing different optimisers, VGG16 and VGG19 consistently demonstrated inferior performance. The research introduced a novel approach utilising the DenseNet201 model and the GWO optimiser for asphalt pavement crack classification, validated against various CNN models. Its robustness was verified by testing against images contaminated with differing levels and types of noise, yielding promising outcomes. Results underscore the method's potential for accurately detecting diverse crack types, implying applicability in real-world scenarios.

1. Introduction

Due to recent global extreme weather, asphalt pavement deterioration, such as cracks, has developed drastically. This has negatively impacted pavement service performance and service life [1]. Detecting cracks early and planning timely maintenance measures could reduce maintenance costs [2]. The demand for a proficient and expeditious automated detection method capable of substituting human-conducted surveys arises from manual detection approaches' laborious and inefficient characteristics that heavily depend on subjective evaluations made by surveyors. Cracks are early signs of asphalt pavement deterioration, and their detection and classification are significant for maintenance strategy and decision-making [3].

The rapid progress of artificial intelligence (AI) has led to a growing interest in deep learning (DL)-based crack detection as a possible solution for the issues associated with manual inspection. Convolutional neural networks (CNNs) are commonly used in computer vision for deep learning. CNN has demonstrated exceptional performance and

competitiveness in several domains, including image classification, semantic segmentation, and object identification [4]. Various studies have found that CNN is becoming a more widely used and effective technique for detecting cracks [5–9].

However, studies using CNNs generally need huge amounts of well-collected data to train a network. Underfitting will probably occur if sufficient training data leads to accurate results [10]. To avoid any possible underfitting, transfer learning methods focus on transferring knowledge from earlier trained CNNs [11].

Pre-trained models can be leveraged in transfer learning techniques to speed up training times using the weights previously trained on the ImageNet dataset. The versatility of this technique lies in its ability to seamlessly integrate pre-trained networks into feature extraction pre-processing and the development of new models. As a result, it effectively reduces training duration and minimises generalisation errors [12].

Various studies have utilised pre-trained CNN models to address the challenge of crack identification and classification in pavement. Qayyum, et al. [13] evaluated seven pre-trained networks, specifically

* Corresponding authors.

E-mail addresses: f.rahimian@tees.ac.uk (F. Pour Rahimian), sepehr.abrishami@port.ac.uk (S. Abrishami).

<https://doi.org/10.1016/j.autcon.2024.105297>

Received 31 October 2023; Received in revised form 22 December 2023; Accepted 24 January 2024

Available online 31 January 2024

0926-5805/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

GoogLeNet, Inception-V3, ResNet18, 50, 101, MobileNet-V2, and ShuffleNet, to identify and classify cracks. The study's findings indicated that Inception-V3 outperformed the other models. Additionally, it was observed that ResNet101 necessitated 171 min for training, but ResNet18 only required 32 min. Hammouch, et al. [14] used a pre-trained VGG19 to detect and categorise road surface defects in Morocco. The study revealed that the F1 score for alligator cracks was 89.34%, while the score for longitudinal cracks was 86.26%. Although the dataset used in this study was relatively large, the authors concluded that a larger set would increase the detection precision.

This contradicts the purpose of using pre-trained models, which have the benefit of not needing as much data as building a model from scratch. Ullah, et al. [15] used three pre-trained models, AlexNet, ResNet18 and SqueezeNet, for crack detection and classification. The study results indicated that AlexNet performs more efficiently and outperforms the other models. AlexNet achieves an F1 score of approximately 97% across different classes, while its overall F1 score is 85.82%. Jana, et al. [16] concluded that GoogleNet outperformed AlexNet and ResNet101 with a high accuracy rate (100%) and zero loss. In a study by Khan and Ahmed [17], a comparison was made between the AlexNet, GoogLeNet, and ResNet18 models. The results indicated that ResNet18 exhibited higher performance in terms of surface condition detection, achieving an overall detection accuracy of 99%. Elghaish, et al. [18] found that the pre-trained models VGG16, VGG19, AlexNet, and GoogleNet had comparable performance in the identification and classification of pavement cracks. Notably, all three models attained accuracy levels above the average accuracy rate of 97.72%.

Hu, et al. [19] conducted experiments on several models from the YOLOv5 series. Their findings indicated that YOLOv5l exhibited the best detection accuracy, with 88.1%. Additionally, YOLOv5s had the quickest detection time, with a mere 11.1 ms per image. Qu, et al. [20] optimised the VGG16 model to enhance its performance in pavement crack identification. The findings of their study revealed that the improved model exhibited a high level of precision in identifying the boundaries of cracks, enhancing the overall accuracy of crack identification. In their study, Dung and Anh [21] utilised the VGG-16 architecture as a foundational network for extracting features, which were subsequently fed into a Softmax classifier. The study results revealed that the accuracy of road crack detection reached 97.8%. Yu et al. [22] presented a hybrid approach integrating CNNs, transfer learning, and data fusion to enhance accuracy and robustness in detecting concrete surface cracks. By effectively fusing the diagnosis results of concrete surfaces from various CNN models, the proposed MDSFA resolves the issue of contradictory prediction results from various networks. The proposed fusion algorithm achieved a BPA of 0.9555, representing a significant enhancement in the confidence level associated with crack detection. Several studies have used transfer learning to develop cracking detection classifiers: MobileNet [23,24], VGG [25,26], GoogleNet [18,27], U-Net [28,29], Res-Net [30,31], Inception [32,33], AlexNet [34,35], YOLO [36,37], and EfficientNet [38]. The aforementioned body of literature affirms that developing a methodology that optimises the precision of the DL model is an ongoing and extensively researched topic.

However, the possibility of using pre-trained models for detecting and classifying different cracks is enormous, in terms of increasing the efficiency and reliability of the crack detection process. Yet, it needs a focus on reducing the dimensionality (reducing the number of features in a dataset) to eliminate unnecessary attributes without harming the learner's performance. Consequently, several feature optimisation algorithms have been employed recently to increase the accuracy of neural networks. Poojary and Pai, [39] utilised three optimisation algorithms, including stochastic gradient descent (SGD), Adam, and RMS Prop, to train popular convolutional neural network (CNN) architectures, specifically ResNet50 and InceptionV3.

In this study, ten pre-trained convolutional neural network (CNN) architectures were compared comprehensively to evaluate their

effectiveness in detecting and classifying cracks in asphalt pavement images. Eight optimisers were employed to assess their impact on the selected ten CNN models. The evaluated pre-trained CNN architectures included AlexNet, VGG16, VGG19, GoogleNet, ResNet101, ShuffleNet, InceptionV3, DenseNet201, DarkNet19, and Xception. Meanwhile, the tested optimizers comprised "principal component analysis" (PCA), "independent component analysis" (ICA), "sparse filtering, feature transformation", "particle swarm optimization" (PSO), "grey wolf optimizer" (GWO), "discrete cosine transforms" (DCT), and "discrete wavelet transform" (DWT).

The assessment of pre-trained models, both pre- and post-optimization, primarily focused on accuracy parameters, particularly the F-score, and their convergence time. All optimizers maintained constant default hyperparameters, except for the learning rate, which was set at 0.001, 0.0001, and 0.0001 in different instances. The study's findings indicate that the grey wolf optimizer (GWO) exhibits superior performance when combined with the DenseNet201 model with an accuracy and F-score of 98.73%. Since the advent of Adam in 2014, there has been a wide array of optimisers available for training high-quality deep learning models. Improved optimizers have shown potential for considerably faster performance and a significant impact on final validation accuracy compared to other methods.

While existing studies have typically focused on evaluating a maximum of two optimisers and their impact on the test CNN architecture, this study stands out by uniquely comparing eight optimisers across ten pre-trained CNN architectures. Its aim is to pinpoint the most effective optimiser for pavement crack classification while considering the influence of noise. As a result, this research offers a valuable reference for other researchers seeking to select suitable CNN models and optimizers for various datasets. Furthermore, the hybrid methodology adopted here—first comparing ten CNN models, then implementing eight optimizers, and finally testing the model performance against different noise types—holds promise for application by other researchers exploring different types of cracks. This approach may lead to identifying the most accurate algorithms in various contexts.

The subsequent sections of this work are structured into six separate sections. Section 2 comprehensively reviews the current body of literature and research on the subject matter under consideration. In this section, various studies, theories, and findings are discussed to establish a foundation for the current work. On the other hand, Section 3 outlines the technique utilised in this study. This section details the specific procedures, techniques, and tools utilised to collect and analyse data and the overall approach taken to address the research objectives. The findings are reported in Section 4, while the analysis and interpretation of the results are discussed in Section 5. In conclusion, Section 6 comprehensively summarises the main points discussed throughout the text.

2. Related works

This section thoroughly examines the existing literature on the utilisation of Convolutional Neural Networks (CNNs) and their prospective use in transfer learning (TL) for the automated identification and categorisation of cracks on pavement surfaces.

2.1. Convolutional neural networks for asphalt pavement crack detection: training from scratch

Cracks were traditionally classified with manual inspection, which is time-consuming, dangerous, and quite skill-dependent [4,40]. Therefore, ensuring such classification efficiency and reliability is challenging due to human subjectivity [38]. Moreover, analytical and logical approaches show a classification accuracy of 80 to 90%, which could be more practical despite fast processing [41]. The limitations of traditional approaches have prompted a growing interest in utilising deep learning-based techniques for crack detection and classification. One of these

methods is the convolutional neural network (CNN). The common CNNs excel over traditional methods are generally twofold. In contrast to traditional approaches in image analysis, such as image processing, CNNs do not rely on manual rules and can extract multi-level feature representations [42] automatically. Furthermore, the level of interoperability exhibited by the CNNs is highly enlightening. The technique of visualising the first-layer weights of a trained CNN has become a common method for inferring the learned features of the network, hence providing insights into the training accuracy [40].

Several recent studies have asserted the power and efficiency of CNNs for crack detection and classification [4,6,38,43]. An example of this is the study of [38], where thirteen CNNs are, from scratch, developed, tailored, and evaluated on the built dataset for the detection and classification of asphalt cracks. An additional illustration may be found in the work of Tan et al., [44], where they have implemented a CNN-based version of the YOLOv3 algorithm to autonomously identify defects in sewage pipes. The research focused on improving the network architecture, bounding box estimation, data augmentation techniques, and loss function. The results indicate that the suggested model outperformed Faster R-CNN and YOLOv3. It acquired a mean average precision (mAP) value of 92%, surpassing the accuracy levels attained in prior research endeavours focused on the automated identification of sewage pipe defects. G. Li et al., [45] introduced U-CliqueNet, a variant of CNN-based U-Net with an alternatively updated clique. The purpose of U-CliqueNet is to effectively isolate cracks from the backdrop in tunnel images. The model was trained using a substantial dataset consisting of 50,000 images for training purposes and 10,000 images for testing purposes. The model demonstrated positive results, achieving a mean pixel accuracy (MPA) of 92.25%, a mean intersection over union (MIoU) of 86.96%, a precision of 86.32%, and an F1-score of 83.40%. In another study conducted by J. Dong et al., [46], the utilisation of StyleGAN and a feature fusion model called Road-Seg-CapsNet was explored as a potential solution to address the issue of edge information loss in current segmentation methodologies while dealing with damaged regions. The study findings indicate that the segmentation-developed model could reach a segmentation accuracy of 0.942. Additionally, it was observed that the minimal accuracy for identifying damaged regions using this model was 0.903.

Although CNNs are common and effective in classification, CNNs require a sheer volume of labelled image data when training from scratch, which is time-consuming, and labelling them is costly, with many road authorities needing more technologies for collecting data. Furthermore, designing the CNN architecture to attain optimal outcomes is extremely challenging, as numerous hyperparameters (such as the convolutional quantity, fully connected layers, and pooling) impact the CNNs' efficacy [40]. To tackle these challenges, the principles of TL and fine-tuning prove to be advantageous.

2.2. Pre-trained convolutional neural networks: coalescence with transfer learning

Retraining a pre-trained CNN model using a new dataset is referred to as transfer learning (TL) and fine-tuning. This involves utilising one of the pre-existing CNN models as a starting point and adapting it to the new dataset. A pre-trained model's capacity for feature extraction and learning prediction rules is employed in the TL process. This approach facilitates the creation of an efficient classifier model on a new dataset. Notably, this method is more cost-effective and computationally reasonable when compared to training a network from scratch. Due to the unique feature, the concept of TL and using pre-trained weights has gained a strong foothold in many domains, including asphalt pavement crack detection. In their study, Mandal, et al. [47] introduced a system that utilised TL to improve the performance of YOLO and accelerate training on road defects such as cracks. The system was built upon the YOLO v2 deep learning framework. Bang, et al. [48] presented a developed model based on TL to identify pavement deterioration in the

pavement surface images. The model used a deep residual network for this purpose. 71 images were used for training the network, sourced from the Crack Forest dataset. Subsequently, the network was tested on a separate collection of 47 images from the same dataset. The deep residual network is utilised to classify all individual pixels inside road images as cracks or backgrounds. In order to optimise efficiency, the utilisation of small training images was employed in TL after adjusting the weights of pre-trained CNNs.

The model suggested by Nie and Wang [49] utilises TL and CNN architectures to detect and classify road crack images. The developed model's performance was analysed concerning the learning technique and regular term coefficient. The proposed pavement crack detection model leverages the network architecture of Faster R-CNN to achieve improved performance. The existing model demonstrates adequate efficacy when applied to pavement images that exhibit little damage and possess a simple background. Ali, et al. [50] presented an automated image-based crack detection method for detecting pavement cracks leveraging CNN. The data set was collected using an unmanned vehicle to train and validate the proposed system. The presented method projects a validation and test accuracy of 92% and 90%, respectively.

Liu, et al. [8] presented U-Net for concrete crack detection. With high efficacy, the trained U-Net could detect cracks in raw images under different conditions (e.g., brightness, disturbed background, etc.). The model's precision, developed by training on a dataset of 57 photos, has a high level of accuracy and can reach 0.9 even in complex cases. Using TL methodology on the VGG-16 pre-trained network resulted in a 94% accuracy rate. By making minor adjustments to a highly trained, fully convolutional layer inside the VGG-16 network, the accuracy rate improved to 98% [21]. The VGG-16 network was fine-tuned by modifying its architecture by lowering the number of completely connected layers from three to two. Additionally, the final classification layer for surface crack detection was adjusted to consist of a Softmax layer with seven tags. The suggested model's mean accuracy of 83.76% outperformed AlexNet, GoogLeNet, and ResNet regarding detection results [51].

They were employing an improved form of a Faster R-CNN known as RetinaNet, Tran, et al. [52] labelled and trained asphalt images. The study's outcomes reported that the trained network realised an accuracy rate of 84.9% in detecting and classifying both crack type and severity level. DenseNet and a deconvolution network architecture were used by Wu, et al. [27] to perform pixel-level detection and fuse features. Crack characteristics in the complicated fine-grained backgrounds of asphalt pavement were learned by utilising a fully convolutional network that incorporated convolutional kernels of various sizes. The findings of the study demonstrate that the suggested methodology achieved a highly effective segmentation outcome for twelve distinct categories of cracks. Moreover, the segmentation of cracks in asphalt pavement has shown significant improvement compared to the most sophisticated techniques currently available. Wu, et al. [33] developed an approach based on a CNN using the pre-trained GoogLeNet Inception V3 for crack identification. Experimental results demonstrated that the trained model's accuracy rate on the test data set reached 98.5%.

Another showcase study conducted [51] presented a pixel-level segmentation CNN called DeepLab_v3+ to segment cracks and calculate physical properties (length and width) in segmentation results. The results demonstrate that DeepLab_v3+ attained accuracy and F-score values of 0.80%, 97.5%, and 0.78%, respectively. Fan and Chung [53] developed a deep residual CNN (Parallel ResNet) in order to create a highly accurate pavement crack detection and monitoring system. The Parallel ResNet achieved the highest results in accuracy (94.27%), recall (92.52%), and F-score value (93.08%). Li, et al. [54] performed a study whereby they devised a system for detecting pavement cracks using vision-based techniques. The primary aim of their research was to develop a pavement surface condition index (PCI) specific to a designated pavement region. In order to categorise the types of cracks present in 5000 pavement distress images, the authors employed a CNN method

together with a genetic algorithm (GA) to enhance the accuracy of the model. The classification accuracy for various types of cracks achieved a notable level of 98%, coupled with a rapid image processing time of 0.047 s. In order to create 12 distinct attention models to identify cracking, Yao, et al. [55] developed a model for pavement crack detection by including the YOLOv5 framework, the spatial and channel squeeze and excitation (SCSE) module, and the convolutional block attention module (CBAM). The study results revealed that the network had a processing speed of 13.15 milliseconds per image while attaining an accuracy rate of 94.4%.

Qu, et al. [28] conducted another study in which a CNN and a transformer named CrackT-net were proposed for crack segmentation. The authors used richer features (RF) UNet++ and polarised self-attention to enhance feature representation capabilities. In addition, they replaced the last feature extraction layer by the transformer. Results showed that the proposed method proved effective, with F-score values of 0.856, 0.700 and 0.637 on three data sets. Huyan, et al. [56] aimed to address the existing knowledge gap by bridging the gap between powerful DL techniques and less effective pixel-level crack identification systems. Two architectures, namely VGGCrackU-Net and ResCrackU-Net, were developed using asphalt pavement crack images gathered from various sources, such as cell phones, action cameras, and autonomous pavement monitoring systems. The images were obtained from different functional classes of pavements. The authors asserted that the created models exhibit benefits compared to the performance of FCN Net and PSPNet. Using CNN, Hammouch, et al. [14] developed an automated approach for the detection and classification of pavement cracks. By implementing TL through the evaluation of a pre-trained VGG-19 model, they could detect and classify cracks effectively. A CNN-based algorithm for the per-pixel labelling of pavement cracks was proposed by Doğan and Ergen [57]. Techniques for residual connections were implemented to increase the robustness of the model.

Several studies have recently compared various pre-trained CNNs and reported different results. For example, Wang and Guo [58] developed models for detecting full convolutional instance segmentation (FCIS) by training them on the transfer learning principle using three cutting-edge CNNs—VGG-16, ResNet-101, and a feature pyramid network (FPN)—as feature extractors and a Faster R-CNN as the backbone. ResNet exhibited the most precise performance with the least training cost among the three models. The performance of the FPN was close to that of ResNet, except for a recall advantage. Several pre-trained models, namely Inception-v3, AlexNet, GoogLeNet, SqueezeNet, ResNet-18, ResNet-50, ResNet-101, and DenseNet-201, were retrained using transfer learning on pavement images. In general, performance metrics indicate that SqueezeNet and GoogleNet outperform the alternative models [59]. Elghaish, et al. [60] developed a CNN model for detecting and classifying highway cracks by testing four pre-trained CNN models (i.e., AlexNet, VGG-16, VGG-19 and GoogLeNet). The findings revealed that the accuracies of every pre-trained model exceeded the average and calculated accuracies for AlexNet and GoogleNet models by >5% (97.72%). Guzmán-Torres, et al. [61] conducted experiments using several CNN models to identify the most effective model for detecting concrete cracks. The authors' analysis led them to propose a novel transfer learning technique for enhancing the precision of deep learning classifiers with micro- and macro-cracks. The architecture of the implemented technique was derived from an enhanced version of the VGG-16 model, resulting in an accuracy of 99.5% and an F-score value of 100%.

2.3. Optimised convolutional neural networks- feature selection Optimisers

Several studies compared the performance of various optimisers using different CNN architectures; their approaches vary depending on the CNN architecture, datasets, and optimiser type. A CNN trained on the "big data" ImageNet database was utilised by Gopalakrishnan, et al.

[5] to automatically detect hot mix bitumen cracks. The performance of several machine learning classifiers was evaluated, and it was found that the single-layer neural network classifier, which utilised the 'Adam' optimiser and was trained on ImageNet pre-trained VGG-16 CNN features, exhibited the highest level of performance. Fine-tuning the last convolutional blocks of the CNN architecture in combination with deep TL is another proven method to increase pavement crack detection accuracy. In research conducted by Chen, et al. [62], it was shown that the utilisation of the Adam optimiser and batch normalisation (BN) method resulted in accelerated convergence of the model and the attainment of a peak accuracy of 99.71%. Dimensionality reduction was achieved using the spider monkey optimisation (SMO) method. A deep neural network (DNN) was then fed with the reduced dataset. The DNN model, based on SMO, produced classification results with accuracies of 99.4% and 92%. Additionally, precision values of 99.5% and 92.7%, recall values of 99.5% and 92.8%, and F1 scores of 99.6% and 92.7% were achieved. Notably, these results were obtained with the minimum training time [63].

Elhariri, et al. [64] examined the effects of employing multi-objective optimisation techniques with a hybrid filter-wrapper approach for feature selection in the context of crack severity identification. An improved iteration of the Grey Wolf Optimisation (GWO) algorithm was introduced by Guernine and Kimour [65] to efficiently investigate a specified space of potentially appropriate CNN architectures while optimising their hyperparameters concurrently. Samma, et al. [66] employed a proficient two-layer optimiser to evolve a pre-trained VGG19 model. The comparison experiment conducted on the two-layer optimiser utilised in this study provides evidence supporting its superior performance compared to well-recognised optimisers such as AOA, WGA, RLMPPO, PSO, and CLPSO in the context of VGG-19 filter selection.

A suggested modification to enhance the trained model's generalisation capabilities involves utilising a modified chicken swarm algorithm (ECSA) for optimising the meta-parameters of the deep convolutional neural network model [67]. The experimental findings indicate that the ECSA algorithm performs superiorly to the CSA and PSO algorithms in optimising the meta-parameters of deep CNNs. Specifically, ECSA demonstrates greater optimisation accuracy and a faster convergence rate. Tian, et al. [68] employed the Particle Swarm Optimisation (PSO) algorithm to automate the process of selecting the most suitable hyperparameters for the Generative Adversarial Network (GAN). The experimental findings have provided evidence of the higher performance of the suggested methodology in comparison to the original GAN method.

Several studies have applied the optimisation of the CNN concept. However, these studies investigated one or two optimisers and reported their impact on the test CNN architecture. This study will evaluate eight optimisers using ten pre-trained CNN architectures to find the best optimiser for pavement crack classification.

3. Methodology

3.1. Data collection

The quality and quantity of the dataset directly impact the efficacy of a trained deep-learning model. Therefore, it is important to gather images in a systematic way to be used in the training of CNN models. The insufficient amount of accessible data for deep learning and machine learning is a major limitation in the advancement of artificial intelligence [69]. To overcome such shortcomings, many researchers used existing public datasets such as the AigleRN dataset [70], CrackTree [71], CFD dataset [72], FHWA/LTPP database [5], German Asphalt Pavement Distress (GAPs) dataset [73], Road Damage Detector dataset [74], CRACK500 dataset [75], etc. Nevertheless, the availability of such datasets is limited, highlighting the need to obtain datasets with varying resolutions, exposure settings, crack intensities, and types. Thus, the

dataset used in this study was collected from different public online sources as follows:

- The CRACK500 dataset has a training set containing 1896 images, a validation set comprising 348 images, and a test set including 1124 images. The validation data will be used to pick the network with the highest performance to mitigate the risk of overfitting during the training phase. Subsequently, the optimal model will be assessed using the test data and additional datasets to evaluate its generalizability. Currently, CRACK500 is the most extensive publicly available pavement crack dataset, with thorough pixel-level annotation. [75].
- The GAP dataset comprises 1969 grayscale images exhibiting several distress classifications, including cracks, potholes, and inlaid patches [73]. A total of 509 images were chosen to evaluate the model's ability to generalise. These images were specifically picked based on the criterion that only image sections containing >1000 pixels were included.
- The CrackTree dataset comprises 206 pavement images, each measuring 800×600 pixels, that exhibit a diverse range of crack forms [71]. The dataset's annotation consists of pixel-wise labels, which may be used directly for model evaluation.

The dataset under study primarily examines longitudinal, horizontal, and diagonal cracks observed on asphalt pavements. Consequently, the images within the dataset have been categorised into three distinct categories, as seen in Table 1.

3.2. Pre-trained convolutional neural networks

For the last decade, CNNs have seen significant advancements, characterised by adopting deeper and wider designs. The CNN model comprises many layers, including convolutional layers, rectified linear unit (ReLU) layers, dropout layers, pooling layers, and dense layers. Despite the encouraging outcomes of CNNs in many classification tasks, determining the optimal CNN architecture for a particular application remains a complex process [76]. The CNN architectures presented in the literature are mostly developed via an iterative process of trial and error or influenced by previous works in the field. Some of the common CNNs are MobileNet [23,24], VGG [5,25,26,61], GoogleNet [18,27,77], U-Net [8,28,29], Res-Net [30,31,78], Inception [32,33,79], AlexNet [34,80,81], YOLO [36,37], and EfficientNet [35,38].

By employing transfer learning, it is possible to construct a CNN with enhanced accuracy by building upon pre-trained deeper models. Transfer learning operates on the principle that a model is trained for a new dataset by employing a set of filters learned from a significantly larger dataset. By retaining pre-learned layers and augmenting the CNN model with new layers, it is possible to develop a new and improved CNN model. This study selects and evaluates ten CNN models for transfer learning, as shown in Fig. 1. These models can be utilised in the development of a classification model that achieves excellent performance with a modest computational cost. Table 2 displays the attributes of the selected models.

These pre-trained models are trained, tested, and evaluated using the typical performance evaluation metrics for crack classification: precision, accuracy, recall, F1-score, and specificity. This initial step shows the difference between the various CNNs and identifies the best model

for crack classification.

3.3. Optimisation

Finding the fully connected-layer architecture of CNNs and some hyperparameters is an optimisation problem. To solve this problem, eight feature selection algorithms were used in this study to optimise the CNNs hyperparameters to achieve the best network performance (see Fig. 1). The performance of each CNN model was evaluated after introducing these optimisers in which the architecture of the fully connected layer and the learning rate came from the optimiser and were compared to find the best approach for crack classification. Table 3 illustrates the method and approach of each optimiser.

4. Results

The results of the crack classification using the pre-trained models are presented in this section. As previously stated, a comparison was made between ten pre-trained models evaluated on the gathered dataset to determine the most effective model for crack detection. Then, as optimisers, we implemented eight feature selection techniques on these pre-trained models. By comparing their performance before and following optimisation, we determined which model and optimiser were most effective at detecting and classifying cracks.

4.1. Evaluation of pre-trained convolutional neural networks

The study endeavoured to identify and classify cracks using a dataset comprising pavement images and ten distinct pre-trained CNN models—AlexNet, VGG16, VGG19, GoogleNet, ResNet101, ShuffleNet, InceptionV3, DenseNet201, DarkNet19, and Xception—which were trained, validated, and evaluated. The performance of each pre-trained CNN model was assessed by comparing F-score, recall, accuracy, precision, and specificity metrics, which are frequently employed to evaluate CNN model performance. Accuracy is a performance parameter that indicates how often the CNN model can accurately detect and classify cracks relative to the total number of iterative detection runs. Fig. 2 shows that DenseNet201 outperformed other models with 94.12% accuracy, then ShuffleNet (94.07%), and ResNet101 (93.83%). DarkNet19 and Xception accuracy was almost the same, with 91.43% and 91.82%, respectively, while InceptionV3 accuracy was 90.49%. The remaining models, AlexNet, VGG16, VGG19 and GoogleNet ranged from 85.14% to 89.09%. The findings of this study indicate that DenseNet201 and ShuffleNet exhibited superior performance compared to other models, achieving accuracies of about 94%.

The specificity parameter indicates the capability of the CNN model to identify a true negative of each available category. Results show that the specificity measures of DenseNet201 and DarkNet19 outperform all other models with 95.24%, followed by ShuffleNet (94.21%), ResNet101 (93.99%) and InceptionV3 (90.61%). The other pre-trained model's specificity measures ranged from 80.28% to 87.29%, as shown in Fig. 2. Precision measures the accuracy of the CNN model in classifying a crack as positive. It reflects the model's reliability in classifying samples as positive. Fig. 1 shows that Xception outperforms all other models with a precision of 97.33%, followed by DenseNet201 (93.2%), ShuffleNet (92.16%) and ResNet101 (91.96%). The recall performance parameter indicates the capability of the CNN model to identify positive samples. There is a significant correlation between recall and the number of positive samples found; the higher the recall, the more positive samples were detected. Results reveal that Xception recorded the highest recall with 97.73%, followed by ShuffleNet (93.22%), DenseNet201 (93.18%), and ResNet101 (92.86%) while the remaining model recall values ranged from 79.45% to 89.18%. F-score is a metric used to evaluate the performance of the CNN models. It combines precision and recall into a single score. Fig. 1 shows that DenseNet201 achieved the highest F-score with 93.19%, followed by ShuffleNet (92.68%),

Table 1
Summary of the study dataset.

Dataset	Longitudinal	Horizontal	Diagonal	Total
Training	367	366	343	1076
Validation	122	116	110	348
Evaluation	255	240	220	715
Total	744	722	673	2139

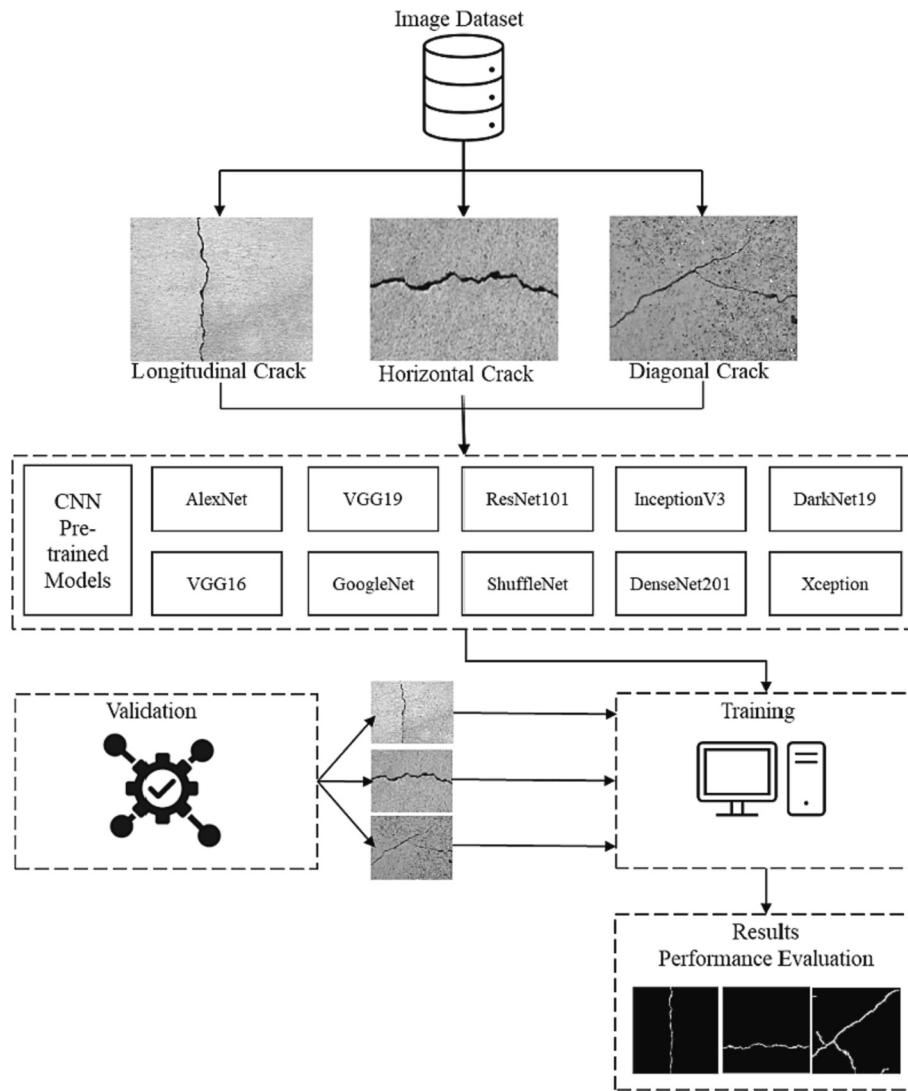


Fig. 1. Flowchart depicting the crack detection and classification process.

Table 2

Comparison between different CNNs in terms of depth, size, and the number of parameters.

CNN	Depth	Size	#Parameters (millions)
AlexNet	8	227 MB	61.0
VGG16	16	515 MB	138
VGG19	19	535 MB	144
GoogleNet	22	27 MB	7.0
ResNet101	101	167 MB	44.6
ShuffleNet	50	5.4 MB	1.4
Inceptionv3	48	89 MB	23.9
DenseNet201	201	77 MB	20.0
DarkNet19	19	78 MB	20.8
Xception	71	85 MB	22.9

Xception (92.44%), ResNet101 (92.4%) and DarkNet19 (91.28%).

The pre-trained performance accuracy parameters are presented and compared in Table 4. Interestingly, the performance accuracy values for all parameters were the lowest for AlexNet, VGG16, VGG19 and GoogleNet compared with the other models. DenseNet201 outperformed all other models, achieving the highest score in all accuracy measures with accuracy (94.12%), specificity (95.24%), precision (93.2%), recall (93.18%), and F-score (93.19%). The fastest model was AlexNet with a processing time (of 2.72 mins), this is due to its smaller depth (8 layers),

while the slowest model was DenseNet201 with a processing time (of 9.42mins) and again, this is because of its larger depth (201 layers).

4.2. Evaluation of the optimised pre-trained convolutional neural networks

Feature selection is one of the most critical components of deep learning, with most real-world data sets having many features. However, not all features are necessary for a specific deep learning model. Utilising many redundant features may lead to several problems, the most significant of which is the computation cost. Due to the unreasonably huge data set, running the model will take an unnecessarily long time. At the same time, it may cause an unexpected overfitting problem. Several feature selection methods are available. The current study's experimental work demonstrates eight popular feature selection methods comprising "principal component analysis" (PCA), "independent component analysis" (ICA), sparse "filtering, feature transformation", "particle swarm optimisation" (PSO), "grey wolf optimiser" (GWO), "discrete cosine transforms" (DCT), and "discrete wavelet transform" (DWT).

Fig. 2 presents a comparison between the eight feature selection optimisers to evaluate the most accurate one and to provide a comprehensive comparison of the accuracy of each feature selection algorithm for each pre-trained CNN model. It can be seen in Fig. 3.1 that the best

Table 3
Feature selection algorithms and methods.

Feature selection	Method
Principal Component Analysis (PCA)	The objective is to decrease the high dimensionality of the detected variables in the data space such that it aligns with the lower intrinsic dimensionality of the independent variables in the feature space.
Independent Component Analysis (ICA)	The process involves identifying latent factors inside a dataset by converting a group of variables into a new set that exhibits maximum independence.
Sparse Filtering	By explicitly maximising the entropy of the learned representation through the maximisation of the proxy of sparsity.
Feature Transformation	It is a mathematical transformation in which a mathematical formula is applied to a particular feature and transforms the values useful for further analysis.
Particle Swarm Optimisation (PSO)	It uses multiple particles that form the swarm. Each particle refers to a candidate solution. The set of candidate solutions coexists and cooperates simultaneously.
Grey Wolf Optimizer (GWO)	The optimisation process involves the implementation of three primary stages in hunting: the search for prey, the encirclement of prey, and the subsequent attack on prey.
Discrete Cosine Transform (DCT)	The given input is transformed into a linear combination using weighted basis functions.
Discrete Wavelet Transform (DWT)	Simultaneously provides spatial and frequency representations of the image.

optimisers for DenseNet201 are GWO and DWT, achieving the highest accuracy of 98.73% and 98.16% and the highest F-score of 98.72% and 98.12, respectively. However, the OpDenseNet201 achieved high accuracy and F-score (>97%) using feature transformation, DCT, ICA, and PCA optimisers. DarkNet19's best performance was recorded using the feature transformation optimiser with an accuracy of 96.97% and an F-score (of 98.81%), followed by DWT with an accuracy of 96.43% and an F-score of 96.51%. DarkNet19's performance accuracy was 91.82% using the Sparse Filtering optimiser, while the F-score was 97.53%. Meanwhile, the PSO optimiser enhanced the model's accuracy to 95.24% and the F-score to 95.34%. The remaining optimisers had minimal impact on the DarkNet19 performance, as shown in Fig. 3.2. In Fig. 3.3, the best optimisers that enhanced the performance of the Xception CNN model are ICT and feature transformation with an accuracy of 96.47% and F-score 95.4%, followed by GWO with an accuracy of 95.91% and F-score of 96%, and PSO with an accuracy of 94.67% and F-score of 94.21%. At the same time, AlexNet's performance was the best using GWO, with an accuracy of 92.83% and an F-score of 92.73%. The impact of the other optimisers was negligible, with accuracy values ranging from 87.27% to 90.96%, as shown in Fig. 3.4.

VGG16 model performance was enhanced slightly using the feature transformation optimiser; the model accuracy was 87.25%, and the F-score was 87.7%, followed by ICA, sparse filtering, and PCA with accuracy and F-score >86% (see Fig. 3.5). Similarly, the performance of the VGG19 model improved slightly using the DWT optimiser with an accuracy of 89.42% and F-score of 88.94%, followed by feature transformation and DCT optimisers with accuracy and F-score >88%, as shown in Fig. 3.6. The best optimisers that enhanced the performance of

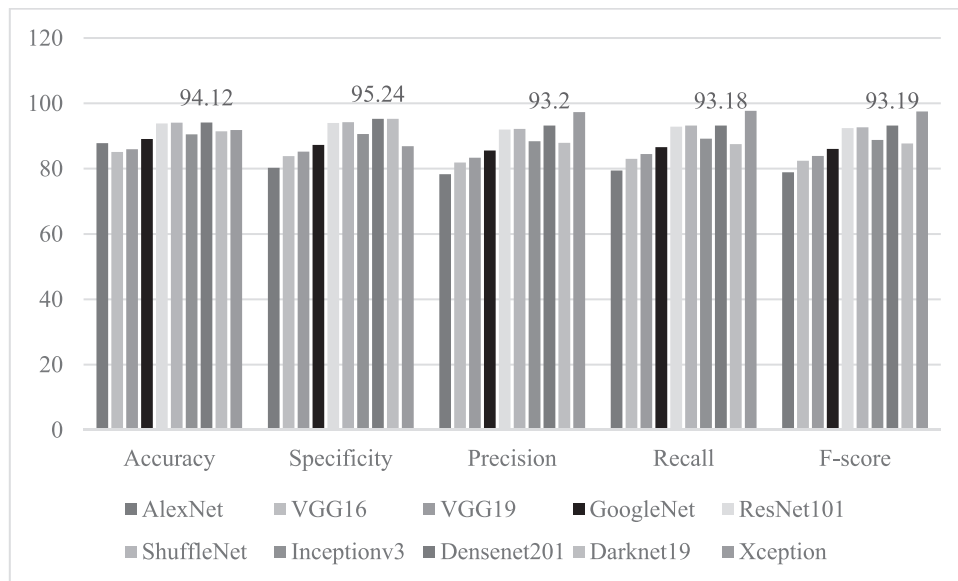


Fig. 2. Graphical comparison between the pre-trained CNNs.

Table 4
The performance of Pre-trained CNNs without applying feature selection techniques.

CNN	Accuracy	Specificity	Precision	Recall	F-score	Time (min)
AlexNet	87.83	80.28	78.32	79.45	78.88	2.72
VGG16	85.14	83.83	81.87	83	82.43	3.99
VGG19	85.93	85.24	83.35	84.46	83.90	5.008
GoogleNet	89.09	87.29	85.56	86.58	86.06	5.35
ResNet101	93.83	93.99	91.96	92.86	92.40	7.31
ShuffleNet	94.07	94.21	92.16	93.22	92.68	6.7
InceptionV3	90.49	90.61	88.42	89.18	88.79	6.51
DenseNet201	94.12	95.24	93.2	93.18	93.19	9.42
DarkNet19	91.43	95.24	87.91	87.50	87.70	5.29
Xception	91.82	86.90	92.33	92.73	92.44	7.86

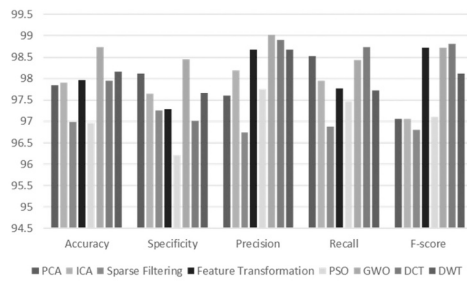


Figure 3.1: Performance of optimized DenseNet201 (OpDenseNet201)

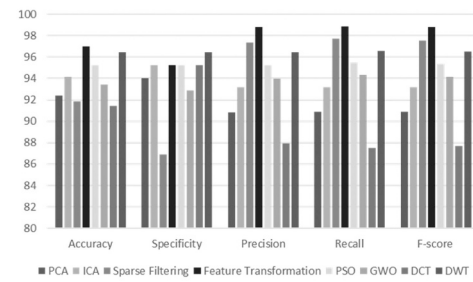


Figure 3.2: Performance of optimized DarkNet19 (OpDarkNet19)

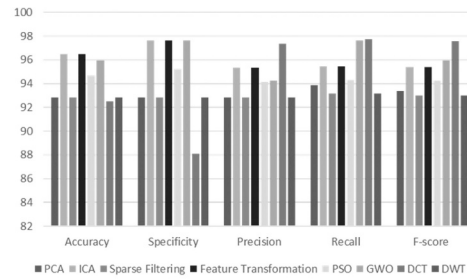


Figure 3.3: Performance of optimized Xception (OpXception)

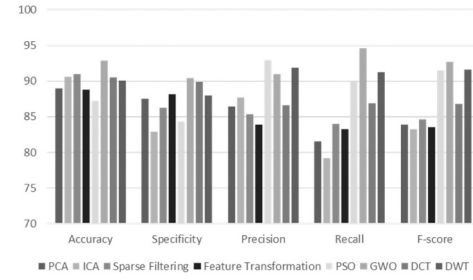


Figure 3.4: Performance of optimized AlexNet (OpAlexNet)

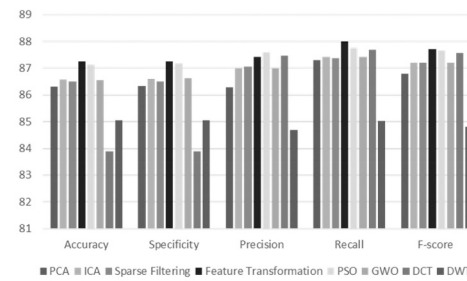


Figure 3.5: Performance of optimized VGG16 (OpVGG16)

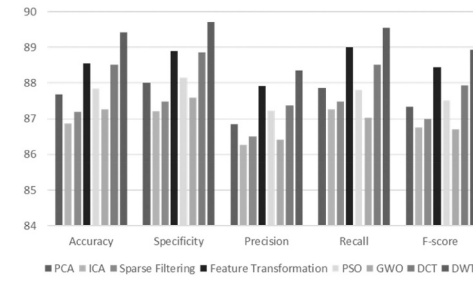


Figure 3.6: Performance of optimized VGG19 (OpVGG19)

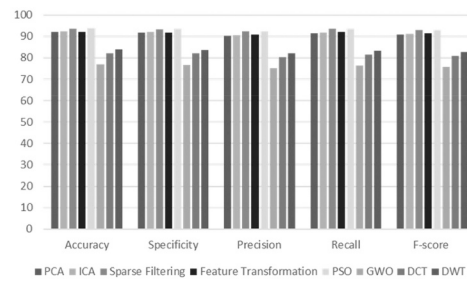


Figure 3.7: Performance of optimized GoogleNet (OpGoogleNet)

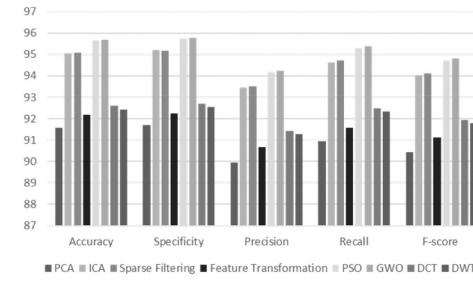


Figure 3.8: Performance of optimized ResNet101 (OpResNet101)

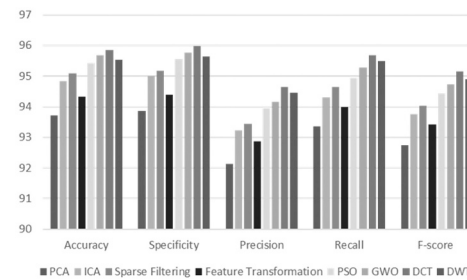


Figure 3.9: Performance of optimized ShuffleNet (OpShuffleNet)

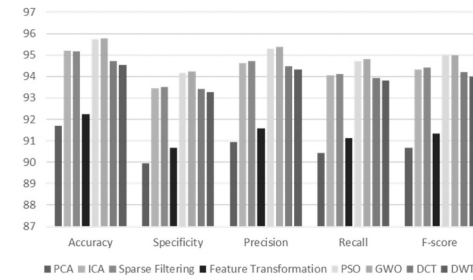


Figure 3.10: Performance of optimized InceptionV3 (OpInceptionV3)

Fig. 3. Performance of optimised pre-trained CNN models.

the GoogleNet model are PSO and Sparse Filtering, with an accuracy of 93.78% and 93.63%, respectively, followed by ICA, feature transformation, and PCA optimisers, with an accuracy of 92.22%, 92.14% and 91.98% as shown in Fig. 3.7. ResNet101 performed with almost the same accuracy value (>95%) and F-score (>94%) using GWO, PSO, ICA and Sparse Filtering optimisers (see Fig. 3.8). ShuffleNet's best optimisers, as shown in Fig. 3.9, are DCT, GWO, DWT, PSO and sparse filtering with accuracy (>95%) and F-score (>94%). Finally, the best optimisers that enhanced the performance of InceptionV3 are GWO, PSO, ICA and sparse filtering with accuracy values (>95%) and F-score (>94%) (see Fig. 3.10).

4.3. Comparison between the proposed optimised pre-trained CNN models and the existing pre-trained CNN models

As shown in Fig. 3, the performance of the optimised pre-trained CNN models varied depending on the applied optimiser. It was observed that while some pre-trained models performed comparatively well across all optimisers, others, such as AlexNet, VGG16, VGG19, GoogleNet, and ResNet101, performed poorly using some optimisers and enhanced slightly with other optimisers. However, the AlexNet model performed excellently using the GWO optimiser (accuracy increased from 87.83% to 92.83%). Similarly, using PSO optimiser, GoogleNet model accuracy increased from 89.09% to 93.78%. The results indicate that it is fundamental to carefully select the best optimisers to enhance the existing pre-trained model's performance to classify pavement cracks. Another noteworthy finding derived from Fig. 4 was the apparent versatility of DenseNet201 as a classifier, both before and after the implementation of optimisers. The existing pre-trained model (DenseNet201) outperformed all other models with an accuracy of 94.12%, and the optimised pre-trained model (OpDenseNet201) outperformed all other models with a high accuracy of 98.73% using the GWO optimiser, as shown in Table 5.

The study's results suggested that the classifier with the highest accuracy and F-score ranking (DenseNet201) performed well using various optimisers. As shown in Fig. 3, DenseNet201 performed well using various optimisers with an accuracy >97% and an F-score >96%. As shown in Table 5, the impact of applying optimisers had a slight improvement on the performance of some existing pre-trained models,

Table 5

Comparison between the proposed optimised pre-trained CNN models and the existing pre-trained CNN models.

CNN	Accuracy	Specificity	Precision	Recall	F-score
AlexNet	87.83	80.28	78.32	79.45	78.88
VGG16	85.14	83.83	81.87	83	82.43
VGG19	85.93	85.24	83.35	84.46	83.90
GoogleNet	89.09	87.29	85.56	86.58	86.06
ResNet101	93.83	93.99	91.96	92.86	92.40
ShuffleNet	94.07	94.21	92.16	93.22	92.68
Inceptionv3	90.49	90.61	88.42	89.18	88.79
DenseNet201	94.12	95.24	93.2	93.18	93.19
DarkNet19	91.43	95.24	87.91	87.50	87.70
Xception	91.82	86.90	92.33	92.73	92.44
OpAlexnet	92.83	90.44	90.93	94.61	92.73
OpVGG16	87.25	87.25	87.41	88.01	87.71
OpVGG19	89.42	89.71	88.35	89.55	88.95
OpGoogleNet	93.78	93.51	92.4	93.6	93.00
OpResNet101	95.69	95.79	94.24	95.37	94.80
OpShuffleNet	95.86	95.98	94.65	95.68	95.16
OpInceptionv3	95.79	94.24	95.37	94.8	95.08
OpDenseNet201	98.73	98.45	99.02	98.44	98.73
OpDarkNet19	96.97	95.24	98.77	98.86	98.82
OpXception	96.47	97.62	95.35	95.45	95.40

such as VGG16, VGG19 and ShuffleNet. Meanwhile, the best pre-trained model was DenseNet201 using GWO optimiser, followed by DarkNet19 and Xception using feature transformation optimiser. Results revealed that the best optimisers across the ten models were GWO (1 in 4) and Feature Transformation (1 in 3). However, these two optimisers took more processing time, approximately 50 s, while other optimisers took less processing time, ranging from 14 s to 29 s.

4.4. The proposed architecture of OpDenseNet201

The schematic diagram of the proposed OpDenseNet201 model is shown in Fig. 5. The proposed model comprises two fundamental stages: firstly, the hyper-parameters are optimised utilising GWO, and secondly, the DenseNet201 network is trained using the optimised hyper-parameters. The utilisation of numerous hyper-parameter annotations by researchers on CNNs led to a multitude of optimisation objectives.

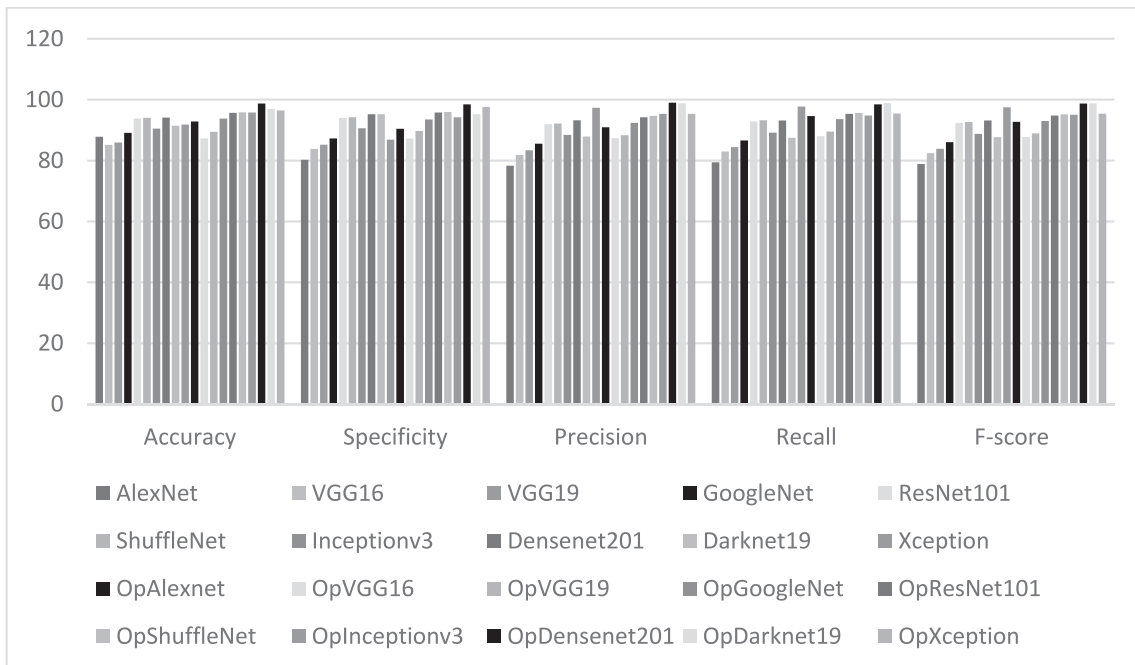


Fig. 4. Graphical comparison between the proposed optimised pre-trained CNN models and the existing pre-trained CNN models.

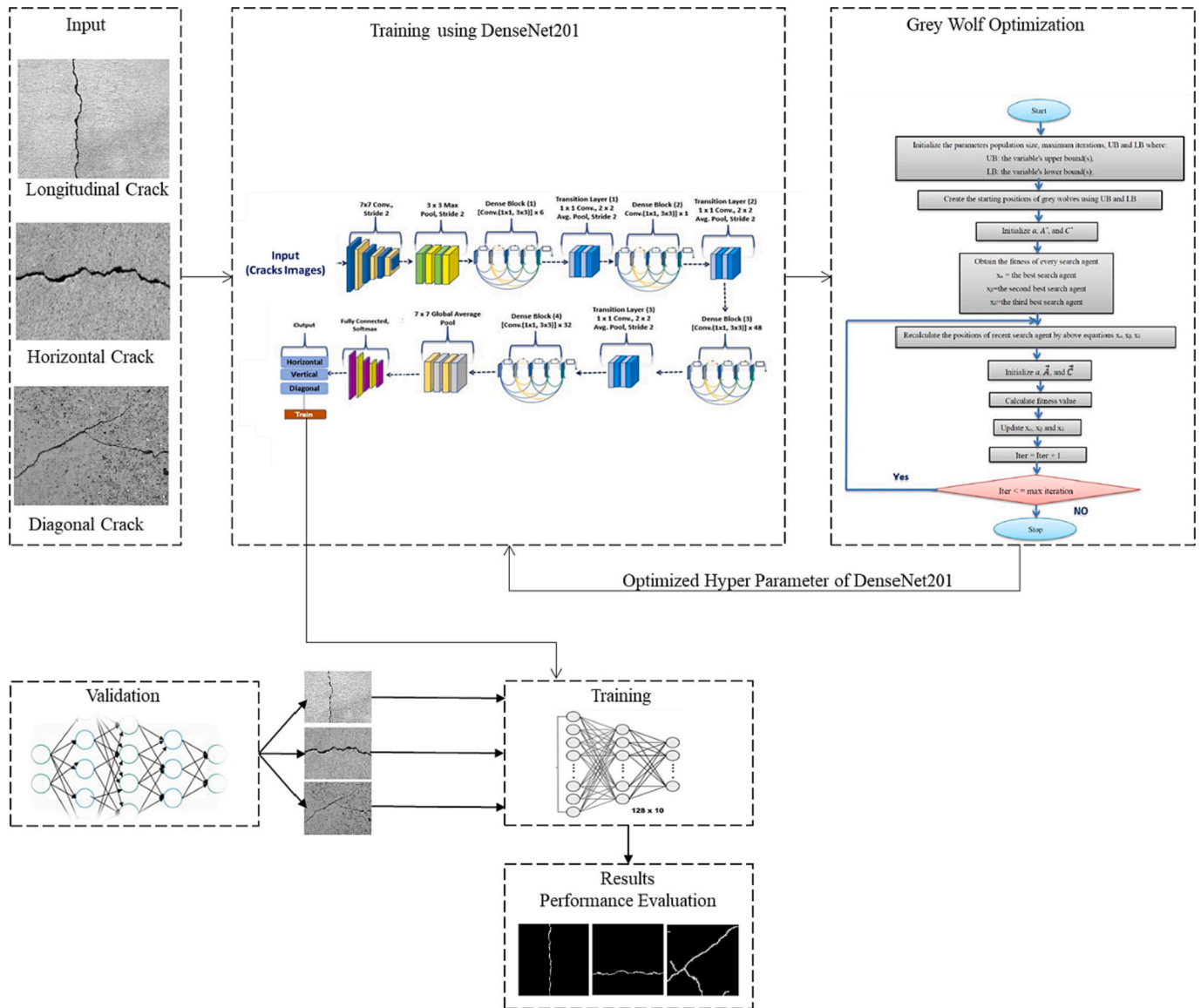


Fig. 5. The proposed OpDenseNet201 schematic diagram for crack classification.

Some take a restricted approach to the problem, considering the hyperparameters of different layers while preserving the network's fixed architecture. An alternative viewpoint regarding hyper-parameters considers factors such as the layer number, learning rate, dropout rate, etc. In the present study, only the earlier is taken into consideration. After DenseNet201 architecture has been selected, its hyper-parameters are optimised through GWO.

The training, which is also employed as the fitness function of the heuristic technique, consumes approximately the most time during the network hyper-parameter optimisation. Some studies trained the model until it converged [82], while others employed chosen data sets [83] to determine the fitness of potential solutions. Using a lightweight model and a small random sample, the DenseNet201 has been trained in this study. Then, any unneeded iterations were discarded by determining if the accuracy change was minimal. The general procedure for the proposed method is illustrated in Fig. 5.

The goal of model training is to minimise the value of the loss function while ensuring that the trained model is neither overfitted nor underfitted for the training patches. To achieve this, the optimisation algorithm (GWO) parameters were tuned based on the convergence progress to adjust their behaviour during runtime. Furthermore,

multiple iterations of the optimisation algorithm with varying initial conditions (mini-batch size and learning rate) were performed to boost the possibilities of discovering a global optimum. As a result, varying the mini-batch size and learning rate along with parameter tuning overcomes the problem of computation memory while simultaneously avoiding the local optimum. Table 6 presents the performance of the

Table 6

Performance of the OpDenseNet model for different values of mini-batch size and learning rate.

Mini-Batch Size	Learning Rate	Accuracy	Specificity	Precision	Recall	F-score
32	0.01	97.83	96.99	96.96	94.86	95.89
	0.001	98.07	97.21	95.16	93.22	94.18
	0.0001	98	96.13	97.34	96.53	96.93
	0.01	98.48	96.62	96.82	95.78	96.29
64	0.001	98.73	98.45	99.02	98.44	98.73
	0.0001	98.1	97.28	97.99	96.34	97.16
Mean		98.2	97.11	97.2	95.8	96.5
Variance		0.112	0.6084	1.66	3.0621	2.277
Standard Deviation		0.335	0.78	1.289	1.7499	1.509

OpDenseNet model for different values of mini-batch size and learning rate.

4.5. The performance of OpDenseNet201 in adverse environmental conditions

To investigate the resilience and flexibility of the OpDenseNet201 network, this study employs 180 images (60 images per label) from the original testing set to assess further the network's classification performance in challenging environmental conditions. The five types of noise included in the selection process are salt & pepper, Gaussian, Speckle, Poisson, and motion blur, as shown in Fig. 6 and defined as follows:

- Impulse noise, or salt and pepper noise, is a form of image noise characterised by the arbitrary occurrence of white and black pixels. The name “salt and pepper” is established through the visual representation of the dispersed black and white particles (salt and pepper, respectively) across the image. In this study, salt & pepper noise is considered to have an intensity of 0.2.
- Gaussian noise, alternatively referred to as Gaussian white noise, is a category of statistical noise characterised by its adherence to a Gaussian or normal distribution. In this study, Gaussian noise is considered with a variance of 0.2.
- Speckle noise is a prevalent form of noise that manifests as an agranular effect in images. Grainy appearance results from stochastic fluctuations in pixel intensity, which define this occurrence. This analysis incorporates speckle noise, which has a variance of 0.2.

- Poisson noise, a form of statistical noise, is generated when counting discrete events within a fixed interval of time or space involves inherent randomisation.
- A motion haze occurs when an image loses its clarity or sharpness. This investigation utilises the coordinates (21,11) and “len, θ ,” where “len” denotes the extent of motion and “ θ ” signifies the angle of motion observed in the anti-clockwise direction.

The analysis and evaluation of the five image categories using the common accuracy metrics are depicted in Table 7; the accuracy metrics of the network are mainly affected by the blurred images, resulting in an accuracy of 94.5% and an F-score of 95.44%. On the other hand, the

Table 7

Performance of the proposed **OpDenseNet201** model against various noise sources: salt & pepper, Gaussian, Speckle, Poisson, motion blur, and combined noise sources.

Noise Source	Accuracy	Specificity	Precision	Recall	F-score
Salt & Pepper (0.2)	95.4	93.01	93.1	97.83	95.41
Gaussian (0.2)	96.1	93.91	94.3	99.29	96.73
Speckle (0.2)	95.4	94.55	94.83	99.55	97.13
Poisson	96.4	93.36	93.67	99.64	96.56
Motion Blur (21, 11)	94.5	92.79	93.22	97.78	95.44
Combined Noise Sources	93.4	92.49	92.86	98.26	95.48

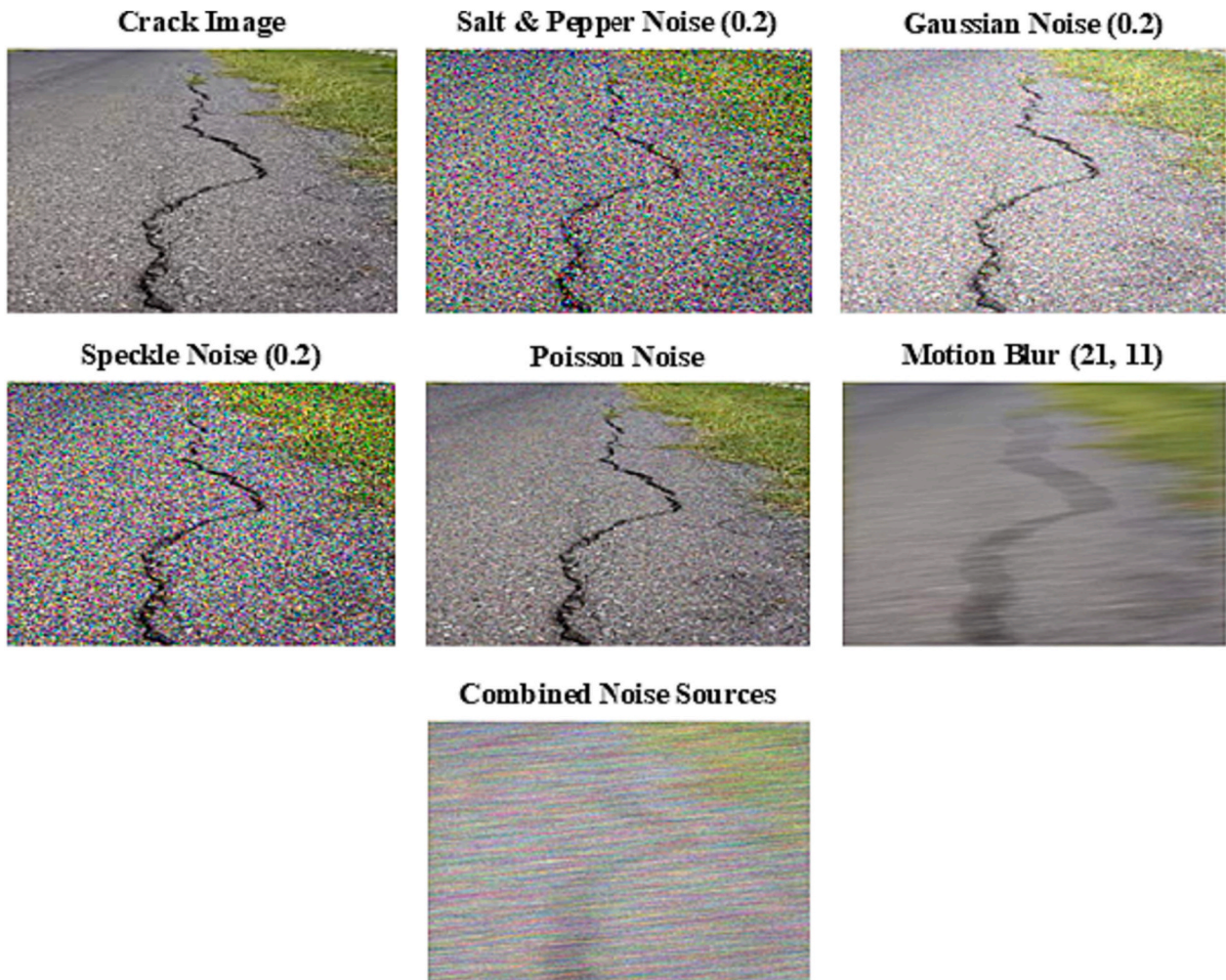


Fig. 6. Example of a crack image polluted with different noise sources.

impact of Gaussian and Poisson is minimal, with the network achieving accuracies of 96.1% and 96.4%, respectively, and F-scores of 96.73% and 96.56%, respectively. The confusion matrices depicted in Fig. 7 demonstrate that all types of noise impact the network's accuracy, with the most detrimental scenario occurring when all noises are combined. The network's overall classification accuracy for 180 testing images is lowered to 93.4% and the F-score to 95.48% due to the combined impact of the five noises. Moreover, Fig. 7 shows that the network accurately classifies the vertical cracks in different noise backgrounds with high true positive values. The horizontal cracks had the highest accuracy when testing the combination of all noise sources, with a total true positive of 169.

To summarise, the OpDenseNet201 network's classification results maintain a high level of consistency even when faced with different background noises. The accuracy and F1 score consistently exceed 93%. These findings demonstrate that the OpDenseNet201 network has strong robustness and adaptability in the presence of background noise.

5. Reflection on results

To detect and classify pavement cracks, ten distinct pre-trained CNN models were trained, evaluated, and validated in this study on an image dataset. The performance of every pre-trained model was evaluated utilising the F-score, accuracy, precision, specificity, and recall, all common performance accuracy parameters. The best performance models were DenseNet201, ShuffleNet and ResNet101, with accuracies of 94.12%, 94.07% and 93.83%, respectively and F-scores of 93.19%, 92.68% and 92.40%, respectively, as shown in Table 4. Eight feature selection techniques were then applied to each pre-trained model to improve the accuracy of crack classification. Following the implementation of several optimisers, the pre-existing models performed training, testing, and validation processes utilising the same dataset. The same performance accuracy parameters were used to select the best pre-trained model and optimiser. Results revealed that not all pre-trained models' performance improved using different optimisers (e.g., VGG19 and VGG16). In comparison, the best optimisers were GWO and feature



Fig. 7. Confusion matrices of the proposed OpDenseNet201 model against various noise sources; salt & pepper, Gaussian, Speckle, Poisson, motion blur, and combined noise sources.

transformation. Meanwhile, the best optimised pre-trained model was DenseNet201 using GWO optimiser with accuracy and an F-score of 98.73%.

Recent studies have presented using pre-trained CNNs in crack identification and classification. Few studies have considered using feature selection algorithms as optimisers [64,66,84,85]. Table 8 summarises several studies that used TL techniques to detect and classify cracks. While the utilisation of TL techniques has demonstrated significant achievements in crack identification and classification, a comprehensive examination of the available literature reveals several limitations. Specifically, only a limited number of research studies have focused on evaluating numerous pre-trained CNNs. A thorough analysis of the performance of various pre-trained CNN models can yield valuable insights into the efficacy of such models in achieving accurate classification.

Moreover, very few prior studies considered optimising the pre-trained CNN models. Most existing studies considered using one or two optimisers to optimise one CNN model. A comprehensive analysis of various feature selection optimisers can offer an in-depth understanding of the potential benefits of employing a well-tuned model with a feature selection optimiser in constructing an automated transfer learning-based crack categorisation model. Thus, this study investigated the performance of ten pre-trained CNNs using eight optimisers to present the well-tuned pre-trained CNN model for asphalt crack classification.

The proposed optimised OpDenseNet201 model was tested across various epochs to evaluate its performance. Fig. 8 illustrates that the optimised model consistently demonstrated high accuracy, precision, and F-score for all epochs. The highest F-score was achieved at epoch 10,000, recording 99.1. Therefore, utilising the proposed optimised OpDenseNet201 model at epoch 10,000 is recommended.

6. Conclusion

This study adopted transfer learning techniques to retrain ten CNN models to classify 2139 images into three pavement crack labels: longitudinal, horizontal, and diagonal. In addition, eight feature selection techniques were applied to each pre-trained CNN model to improve fairly accurate pavement crack classification. The following findings are based on the study's particular transfer learning techniques:

- (1) The results reveal that the performance of some pre-trained CNN models is better than others for pavement crack classification. For example, DenseNet201 was the best CNN model, with an overall accuracy of 94.12%, followed by ShuffleNet and ResNet101, with overall accuracy of 94.07% and 93.83%, respectively. VGG16 was the least accurate of all the models, with an overall accuracy of 85.14%.
- (2) The best-performing pre-trained CNN model after applying different feature selection techniques as optimisers was DenseNet201, with overall accuracy ranging from 96.97% to 98.73% across the eight optimisers. The second-best performing model was DarkNet19, with overall accuracy ranging from 91.43% to 96.97%, followed by Xception, with overall accuracy ranging from 92.5% to 96.47%. VGG16 and VGG19 performed poorly even after applying different optimisers, with overall accuracy ranging from 83.89% to 89.42%.
- (3) The best-performing pre-trained model was DenseNet201 using GWO optimiser with an overall accuracy of 98.73%, followed by DarkNet19 and Xception using feature transformation optimiser with an overall accuracy of 96.97% and 96.47%, respectively.
- (4) The performance of DarkNet19 and Xception improved notably after applying the feature selection techniques. The model's

Table 8

Comparison between the proposed OpDenseNet201 model and the existing state-of-the-art CNN models.

Author	Method	Transfer learning	Deep learning from scratch	CNN optimisation	Results
Qayyum, et al. [13]	Using seven pre-trained CNNs for crack classification	GoogLeNet, Inception-V3, ResNet18, 50 and 101 MobileNet-V2, and ShuffleNet	×	×	The Inception-V3 model performs better than previous models, with an accuracy rate of 96%.
Elghaish, et al. [84]	Using four pre-trained CNN models to classify and detect cracks in highways	AlexNet, VGG16, VGG19, GoogleNet	×	SGDM, Rmsprop and Adam optimisers	Adam's optimisation method attained the highest level of accuracy, reaching 79.62%, while using a learning rate of 0.001.
Samma, et al. [86]	Using a pre-trained CNN for road damage detection	VGG-19	×	Two-layers optimiser	The model that was suggested attained an accuracy of 96.4% in terms of F1-score, while simultaneously reducing the VGG-19 filter by a maximum of 52%.
Elhariri, et al. [64]	The utilisation of pre-trained CNNs to detect cracks and identify their severity.	VGG16	×	Hybrid filter-wrapper and Salp swarm optimisation.	The suggested crack detecting technique demonstrates a general increase in accuracy ranging from about 27% to 57%.
Liu, et al. [38]	Applying deep learning and infrared thermography for asphalt pavement crack severity classification.	MobileNetV2 ResNet-34, ResNet-152 DenseNet-121 DenseNet-161 EfficientNet-B0 -B7	MobileNetV2 ResNet-34 ResNet-152 DenseNet-121 DenseNet-161 EfficientNet-B0 -B7	×	EfficientNet-B3 had the highest accuracy on all three types of images for both deep learning from scratch and transfer learning.
Islam, et al. [87]	Using transfer learning models to detect concrete crack.	VGG16, ResNet18, DenseNet161, and AlexNet	×	×	AlexNet demonstrates superior performance compared to all other examples of performance measures, obtaining an accuracy rate of 99.90%.
Arafin, et al. [88]	To classify concrete cracks	VGG-19, ResNet-50, InceptionV3, Xception, and MobileNetV2	×	SGD	The InceptionV3 model demonstrated superior performance compared to the other models, with an accuracy rate of 91%.
The proposed method	Using transfer learning and feature selection optimisers to classify asphalt cracks	AlexNet VGG16 VGG19 GoogleNet ResNet101 ShuffleNet Inceptionv3 DenseNet201 DarkNet19 Xception	×	PCA, ICA, Sparse Filtering, Feature Transformation, PSO, GWO, DCT and DWT	The best performing pre-trained model was DenseNet201 using GWO optimiser with an overall accuracy of 98.73%, f

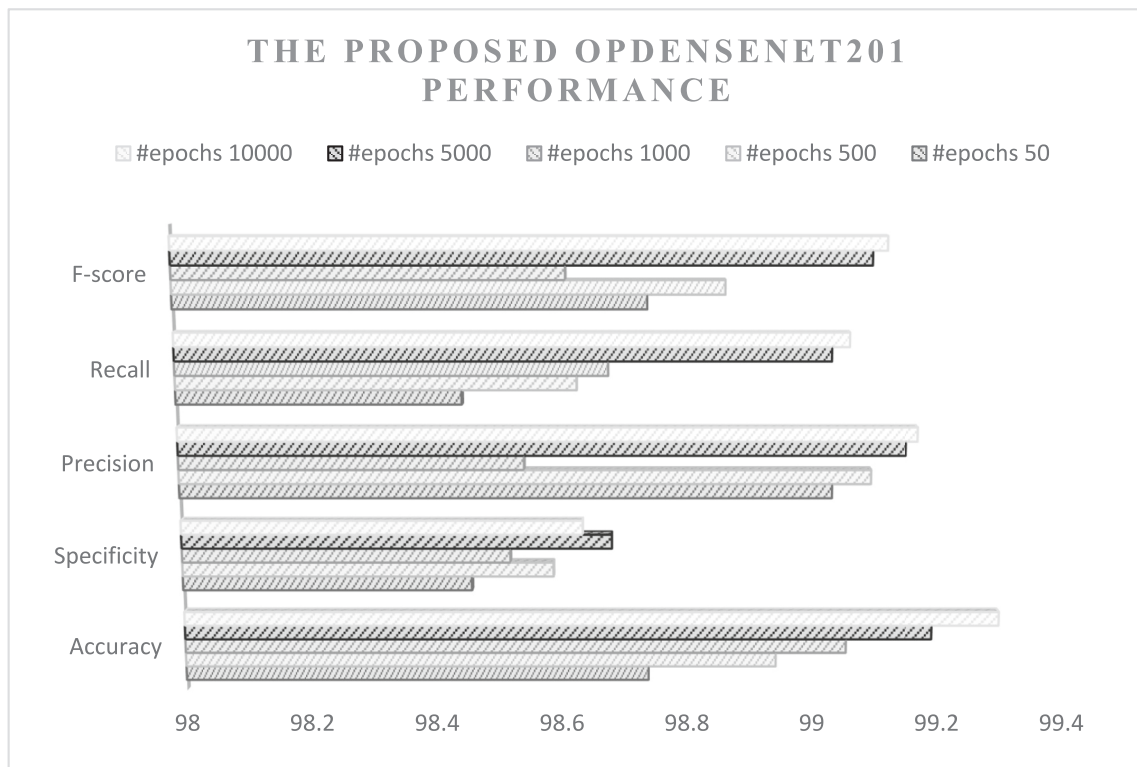


Fig. 8. The proposed OpDenseNet201 performance under different epochs.

accuracy increased from 91.43% to 96.97% and from 91.82% to 96.47%, respectively.

- (5) The best optimisers were GWO and feature transformation across the eight applied optimisers. GWO ranked the first optimiser in 4 models, while feature transformation ranked the first optimiser in 3.
- (6) The study results outlined the proposed method of classifying pavement cracks using the pre-trained model DenseNet201 and GWO optimiser. This proposed method can optimise the crack classification process and classify pavement cracks with high accuracy and optimised running time. Accordingly, the proposed method can reduce the training cost of CNN models from scratch, reducing the cost and hazards of manual inspection and increasing the reliability and efficiency of pavement crack detection.

The study results indicate that the feature selection methods employed in this study as optimisers can be utilised to pre-trained CNN models to improve fairly accurate pavement crack classification. The methodology employed in this study, along with the common performance metrics utilised, will undoubtedly attract the attention of other academics and practitioners contemplating the incorporation of transfer learning methodologies. The results showed that using different feature selection techniques as optimisers could improve performance even for pre-trained models that performed poorly in this study.

In summary, the following are the primary contributions of this paper: (1) A comprehensive assessment of ten distinct pre-trained CNN architectures designed for pavement crack detection and classification; (2) An in-depth examination of eight optimisation techniques concerning the performance of pre-trained CNN models utilised in pavement crack detection and classification; and (3) The construction of an optimised CNN model for crack detection predicated on the most effective pre-trained model. This work focused on only three types of asphalt pavement cracking. In future, more types (e.g., fatigue, edge, alligator, block, and slippage cracking) will be investigated to evaluate the

performance of pre-trained CNN models for asphalt pavement crack detection. Future research should examine the effects of variability in the optimiser selection process to improve the performance and accuracy of the pre-trained CNN model. Finally, future research should examine the performance of the pre-trained CNN models after fine-tuning to identify and classify crack severity by segmenting features such as width and length.

CRediT authorship contribution statement

Sandra Matarneh: Conceptualization, Investigation, Methodology, Project administration, Supervision, Visualization, Writing – original draft. **Faris Elghaish:** Conceptualization, Investigation, Methodology, Supervision, Visualization, Writing – original draft. **Farzad Pour Rahimian:** Conceptualization, Investigation, Methodology, Supervision, Visualization, Writing – review & editing. **Essam Abdellatef:** Methodology, Writing – review & editing. **Sepehr Abrishami:** Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] S. Li, Y. Cao, H. Cai, Automatic pavement-crack detection and segmentation based on steerable matched filtering and an active contour model, *J. Comput. Civ. Eng.* 31 (5) (2017), [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000695](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000695). Art no. 04017045.

- [2] X. Yang, H. Li, Y. Yu, X. Luo, T. Huang, X. Yang, Automatic pixel-level crack detection and measurement using fully convolutional network, *Comput.-Aided Civil Infrastruct. Eng.* 33 (12) (2018) 1090–1109, <https://doi.org/10.1111/mice.12412>.
- [3] A. Cubero-Fernandez, F.J. Rodriguez-Lozano, R. Villatoro, J. Olivares, J. M. Palomares, Efficient pavement crack detection and classification, *Eurasip J. Image Video Process.* 2017 (2017) 1–11, <https://doi.org/10.1186/s13640-017-0187-0>.
- [4] D. Dais, I.E. Bal, E. Smyrou, V. Sarhosis, Automatic crack classification and segmentation on masonry surfaces using convolutional neural networks and transfer learning, *Autom. Constr.* 125 (2021) 103606, <https://doi.org/10.1016/j.autcon.2021.103606>.
- [5] K. Gopalakrishnan, S.K. Khaitan, A. Choudhary, A. Agrawal, Deep convolutional neural networks with transfer learning for computer vision-based data-driven pavement distress detection, *Constr. Build. Mater.* 157 (2017) 322–330, <https://doi.org/10.1016/j.conbuildmat.2017.09.110>, 12/30/ 2017.
- [6] J. Huyen, W. Li, S. Tighe, Z. Xu, J. Zhai, CrackU-net: a novel deep convolutional neural network for pixelwise pavement crack detection, *Struct. Control. Health Monit.* 27 (8) (2020) e2551, <https://doi.org/10.1002/stc.2551>, 8/ / 2020.
- [7] F. Liu, L. Wang, UNet-based model for crack detection integrating visual explanations, *Constr. Build. Mater.* 322 (2022) 126265, <https://doi.org/10.1016/j.conbuildmat.2021.126265>.
- [8] Z. Liu, Y. Cao, Y. Wang, W. Wang, Computer vision-based concrete crack detection using U-net fully convolutional networks, *Autom. Constr.* 104 (2019) 129–139, <https://doi.org/10.1016/j.autcon.2019.04.005>.
- [9] H. Majidifard, Y. Adu-Gyamfi, W.G. Buttlar, Deep machine learning approach to develop a new asphalt pavement condition index, *Constr. Build. Mater.* 247 (2020) 118513, <https://doi.org/10.1016/j.conbuildmat.2020.118513>.
- [10] K. Hongjo, K. Hyoungkwan, H.Y. Won, B. Hyeran, Detecting construction equipment using a region-based fully convolutional network and transfer learning, *J. Comput. Civ. Eng.* 32 (2) (2018) 04017082, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000731](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000731), 3/ / 2018.
- [11] Q. Yang, W. Shi, J. Chen, W. Lin, Deep convolution neural network-based transfer learning method for civil infrastructure crack detection, *Autom. Constr.* 116 (2020) 103199, <https://doi.org/10.1016/j.autcon.2020.103199>, 8/ / 2020.
- [12] L. Ai, B. Zhang, P. Ziehl, A transfer learning approach for acoustic emission zonal localization on steel plate-like structure using numerical simulation and unsupervised domain adaptation, *Mech. Syst. Signal Process.* 192 (2023) 110216, <https://doi.org/10.1016/j.ymssp.2023.110216>, 06/01/ 2023.
- [13] W. Qayyum, R. Ehtisham, A. Bahrami, C. Camp, J. Mir, A. Ahmad, Assessment of convolutional neural network pre-trained models for detection and orientation of cracks, *Materials* 16 (2) (2023) 826 [Online]. Available: <https://www.mdpi.com/1996-1944/16/2/826>.
- [14] W. Hammouch, C. Chouiekh, G. Khaissidi, M. Mrabti, Crack detection and classification in Moroccan pavement using convolutional neural network, *Infrastructures* 7 (11) (2022) 152 [Online]. Available: <https://www.mdpi.com/2412-3811/7/11/152>.
- [15] A. Ullah, H. Elahi, Z. Sun, A. Khatoun, I. Ahmad, Comparative analysis of AlexNet, ResNet18 and SqueezeNet with diverse modification and arduous implementation, *Arab. J. Sci. Eng.* (2022) 1–21, <https://doi.org/10.1007/s13369-021-06182-6>.
- [16] S. Jana, S. Thangam, A. Kishore, V. Sai Kumar, S. Vandana, “transfer learning based deep convolutional neural network model for pavement crack detection from images,” *international journal of nonlinear, Anal. Appl.* 13 (1) (2022) 1209–1223, <https://doi.org/10.22075/IJNAA.2021.24521.2762>.
- [17] M.N. Khan, M.M. Ahmed, Weather and surface condition detection based on road-side webcams: application of pre-trained convolutional neural network, *Int. J. Transport. Sci. Technol.* 11 (3) (2022) 468–483, <https://doi.org/10.1016/j.ijst.2021.06.003>, 09/01/ 2022.
- [18] F. Elghaish, et al., Developing a new deep learning CNN model to detect and classify highway cracks, *J. Eng. Design Technol.* 20 (4) (2022) 993–1014, <https://doi.org/10.1108/JEDT-04-2021-0192>.
- [19] G.X. Hu, B.L. Hu, Z. Yang, L. Huang, P. Li, Pavement crack detection method based on deep learning models, *Wirel. Commun. Mob. Comput.* 2021 (2021) 5573590, <https://doi.org/10.1155/2021/5573590>, 05/17 2021.
- [20] Z. Qu, J. Mei, L. Liu, D.-Y. Zhou, Crack detection of concrete pavement with cross-entropy loss function and improved VGG16 network model, *Ieee Access* 8 (2020) 54564–54573, <https://doi.org/10.1109/ACCESS.2020.2981561>.
- [21] C.V. Dung, L.D. Anh, Autonomous concrete crack detection using deep fully convolutional neural network, *Autom. Constr.* 99 (2019) 52–58, <https://doi.org/10.1016/j.autcon.2018.11.028>, 03/01/ 2019.
- [22] Y. Yu, B. Samali, M. Rashidi, M. Mohammadi, T.N. Nguyen, G. Zhang, Vision-based concrete crack detection using a hybrid framework considering noise effect, *J. Build. Eng.* 61 (2022) 105246, <https://doi.org/10.1016/j.job.2022.105246>, 12/01/ 2022.
- [23] Z.S. Hernanda, H.A. Mahmudah, R.W. Sudibyo, CNN-Based Hyperparameter Optimization Approach for Road Pothole and Crack Detection Systems, in: 2022 IEEE World AI IoT Congress (AllIoT), 2022, pp. 538–543, <https://doi.org/10.1109/AllIoT54504.2022.9817316>.
- [24] Y. Hou, et al., MobileCrack: object classification in asphalt pavements using an adaptive lightweight deep learning, *J. Transportat. Eng. Part B: Pavement.* 147 (1) (2021), [https://doi.org/10.1061/\(JPEODX.0000245](https://doi.org/10.1061/(JPEODX.0000245).
- [25] R. Li, J. Yu, F. Li, R. Yang, Y. Wang, Z. Peng, Automatic bridge crack detection using unmanned aerial vehicle and faster R-CNN, *Constr. Build. Mater.* 362 (2023) 129659, <https://doi.org/10.1016/j.conbuildmat.2022.129659>, 1/ / 2023.
- [26] D.O. Brien, J. Andrew Osborne, E. Perez-Duenas, R. Cunningham, Z. Li, Automated crack classification for the CERN underground tunnel infrastructure using deep learning, *Tunn. Undergr. Space Technol.* 131 (2023) 104668, <https://doi.org/10.1016/J.TUST.2022.104668>, 1/ / 2023.
- [27] Y. Wu, W. Yang, J. Pan, P. Chen, Asphalt pavement crack detection based on multi-scale full convolutional network, *J. Intell. Fuzzy Syst.* 40 (1) (2021) 1495–1508, <https://doi.org/10.3233/JIFS-191105>.
- [28] Z. Qu, Y. Li, Q. Zhou, CrackT-net: a method of convolutional neural network and transformer for crack segmentation, *J. Electro. Imag.* 31 (2) (2022), <https://doi.org/10.1117/1.JEI.31.2.023040>.
- [29] Q. Yang, X. Ji, Automatic pixel-level crack detection for civil infrastructure using Unet++ and deep transfer learning, *IEEE Sensors J.* 21 (17) (2021) 19165–19175, <https://doi.org/10.1109/JSEN.2021.3089718>.
- [30] S.Y. Wang, T. Guo, Transfer learning-based algorithms for the detection of fatigue crack initiation sites: a comparative study, *Front. Mater.* 8 (2021), <https://doi.org/10.3389/fmats.2021.756798>.
- [31] H.Y. Yoon, J.H. Kim, J.W. Jeong, Classification of the sidewalk condition using self-supervised transfer learning for wheelchair safety driving, *Sensors* 22 (1) (2022), <https://doi.org/10.3390/s22010380>.
- [32] L. Ali, F. Alnajjar, H.A. Jassmi, M. Gochoo, W. Khan, M.A. Serhani, Performance evaluation of deep CNN-based crack detection and localization techniques for concrete structures, *Sensors* 21 (5) (2021) 1–22, <https://doi.org/10.3390/s21051688>.
- [33] L. Wu, X. Lin, Z. Chen, P. Lin, S. Cheng, Surface crack detection based on image stitching and transfer learning with pretrained convolutional neural network, *Struct. Control. Health Monit.* 28 (8) (2021), <https://doi.org/10.1002/stc.2766>.
- [34] X. Han, et al., Structural damage-causing concrete cracking detection based on a deep-learning method, *Constr. Build. Mater.* 337 (2022), <https://doi.org/10.1016/j.conbuildmat.2022.127562>.
- [35] F. Liu, J. Liu, L. Wang, Asphalt pavement fatigue crack severity classification by infrared thermography and deep learning, *Autom. Constr.* 143 (2022), <https://doi.org/10.1016/j.autcon.2022.104575>.
- [36] S. Teng, Z. Liu, X. Li, Improved YOLOv3-based bridge surface defect detection by combining high- and low-resolution feature images, *Buildings* 12 (8) (2022), <https://doi.org/10.3390/buildings12081225>.
- [37] J. Liu, Z. Zhao, C. Lv, Y. Ding, H. Chang, Q. Xie, An image enhancement algorithm to improve road tunnel crack transfer detection, *Constr. Build. Mater.* 348 (2022), <https://doi.org/10.1016/j.conbuildmat.2022.128583>.
- [38] F. Liu, J. Liu, L. Wang, Deep learning and infrared thermography for asphalt pavement crack severity classification, *Autom. Constr.* 140 (2022), <https://doi.org/10.1016/j.autcon.2022.104383>.
- [39] R. Poojary, A. Pai, Comparative Study of Model Optimization Techniques in Fine-Tuned CNN Models, in: 2019 International Conference on Electrical and Computing Technologies and Applications (ICECTA), 19–21 Nov. 2019, 2019, pp. 1–4, <https://doi.org/10.1109/ICECTA48151.2019.8959681>.
- [40] F. Liu, J. Liu, L. Wang, Asphalt pavement crack detection based on convolutional neural network and infrared thermography, in: *IEEE Transactions on Intelligent Transportation Systems*, 2022, pp. 1–11, <https://doi.org/10.1109/ITS.2022.3142393>.
- [41] T. Omar, M. Nehdi, Remote sensing of concrete bridge decks using unmanned aerial vehicle infrared thermography, *Autom. Constr.* 07/01 (2017), <https://doi.org/10.1016/j.autcon.2017.06.024>.
- [42] Y. Liu, J. Yao, X. Lu, R. Xie, L. Li, DeepCrack: a deep hierarchical feature learning architecture for crack segmentation, *Neurocomputing* 338 (2019) 139–153, <https://doi.org/10.1016/j.neucom.2019.01.036>.
- [43] X. Liu, L. Zhu, Y. Wang, Z. Yu, A crack detection system of subway tunnel based on image processing, *Measure. Control (United Kingdom)* 55 (3–4) (2022) 164–177, <https://doi.org/10.1177/00202940211062015>.
- [44] Y. Tan, R. Cai, J. Li, P. Chen, M. Wang, Automatic detection of sewer defects based on improved you only look once algorithm, *Autom. Constr.* 131 (2021) 103912, <https://doi.org/10.1016/j.autcon.2021.103912>, 11/01/ 2021.
- [45] G. Li, B. Ma, S. He, X. Ren, Q. Liu, Automatic tunnel crack detection based on U-net and a convolutional neural network with alternately updated clique, *Sensors* 20 (3) (2020), <https://doi.org/10.3390/s20030717>.
- [46] J. Dong, et al., Innovative method for pavement multiple damages segmentation and measurement by the Road-Seg-CapsNet of feature fusion, *Constr. Build. Mater.* 324 (2022), <https://doi.org/10.1016/j.conbuildmat.2022.126719>. Art no. 126719.
- [47] V. Mandal, L. Uong, Y. Adu-Gyamfi, Automated road crack detection using deep convolutional neural networks, in: 2018 IEEE International Conference on Big Data (Big Data), 2018, pp. 5212–5215, <https://doi.org/10.1109/BigData.2018.8622327>.
- [48] S. Bang, S. Park, H. Kim, Y.-S. Yoon, H. Kim, A Deep Residual Network with Transfer Learning for Pixel-level Road Crack Detection, 2018, <https://doi.org/10.22260/ISARC2018/0103>.
- [49] M. Nie, K. Wang, Pavement Distress Detection Based on Transfer Learning, 2018, pp. 435–439, <https://doi.org/10.1109/ICSAI.2018.8599473>.
- [50] L. Ali, N.K. Valappil, D.N.A. Kareem, M.J. John, H.A. Jassmi, Pavement Crack Detection and Localization using Convolutional Neural Networks (CNNs), in: 2019 International Conference on Digitization (ICD), 18–19 Nov. 2019, 2019, pp. 217–221, <https://doi.org/10.1109/ICD47981.2019.9105786>.
- [51] J. Zhu, J. Song, An intelligent classification model for surface defects on cement concrete bridges, *Appl. Sci. (Switzerland)* 10 (3) (2020), <https://doi.org/10.3390/app10030972>.
- [52] V.P. Tran, T.S. Tran, H.J. Lee, K.D. Kim, J. Baek, T.T. Nguyen, One stage detector (RetinaNet)-based crack detection for asphalt pavements considering pavement distresses and surface objects, *J. Civ. Struct. Heal. Monit.* 11 (1) (2021) 205–222, <https://doi.org/10.1007/s13349-020-00447-8>.

- [53] C.-L. Fan, Y.-J. Chung, Design and optimization of CNN architecture to identify the types of damage imagery, *Mathematics* 10 (19) (2022) 3483 [Online]. Available: <https://www.mdpi.com/2227-7390/10/19/3483>.
- [54] J. Li, T. Liu, X. Wang, J. Yu, Automated asphalt pavement damage rate detection based on optimized GA-CNN, *Autom. Constr.* 136 (2022), <https://doi.org/10.1016/j.autcon.2022.104180>, Art no. 104180.
- [55] H. Yao, Y. Liu, X. Li, Z. You, Y. Feng, W. Lu, A detection method for pavement cracks combining object detection and attention mechanism, *IEEE Trans. Intell. Transp. Syst.* (2022), <https://doi.org/10.1109/TITS.2022.3177210>.
- [56] J. Huiyan, T. Ma, W. Li, H. Yang, Z. Xu, Pixelwise asphalt concrete pavement crack detection via deep learning-based semantic segmentation method, *Struct. Control. Health Monit.* 29 (8) (2022) e2974, <https://doi.org/10.1002/stc.2974>.
- [57] G. Dogan, B. Ergen, A new mobile convolutional neural network-based approach for pixel-wise road surface crack detection, *Measurement* 195 (2022) 111119, <https://doi.org/10.1016/j.measurement.2022.111119>, 05/31/ 2022.
- [58] S.Y. Wang, T. Guo, Transfer learning-based algorithms for the detection of fatigue crack initiation sites: a comparative study, *Front. Mater.* 8 (2021) 756798, 11/01, <https://doi.org/10.3389/fmats.2021.756798>.
- [59] S. Ranjbar, F.M. Nejad, H. Zakeri, An image-based system for pavement crack evaluation using transfer learning and wavelet transform, *Int. J. Pavement Res. Technol.* 14 (4) (2021) 437–449, <https://doi.org/10.1007/s42947-020-0098-9>.
- [60] F. Elghaish, et al., Developing a new deep learning CNN model to detect and classify highway cracks, *J. Eng., Design Technol.* (2021), <https://doi.org/10.1108/JEDT-04-2021-0192> ahead-of-print, no. ahead-of-print.
- [61] J.A. Guzmán-Torres, M.Z. Naser, F.J. Domínguez-Mota, Effective medium crack classification on laboratory concrete specimens via competitive machine learning, *Structures* 37 (2022) 858–870, <https://doi.org/10.1016/j.istruc.2022.01.061>.
- [62] K. Chen, A. Yadav, A. Khan, Y. Meng, K. Zhu, Improved crack detection and recognition based on convolutional neural network, *Modell. Simulat. Eng.* 2019 (2019) 8796743, <https://doi.org/10.1155/2019/8796743>, 10/14 2019.
- [63] N. Khare, et al., SMO-DNN: spider monkey optimization and deep neural network hybrid classifier model for intrusion detection, *Electronics* 9 (2020) 692, 04/24, <https://doi.org/10.3390/electronics9040692>.
- [64] E. Elhariri, N. El-Bendary, S.A. Taie, Using hybrid filter-wrapper feature selection with multi-objective improved-Salp optimization for crack severity recognition, *IEEE Access* 8 (2020) 84290–84315, <https://doi.org/10.1109/ACCESS.2020.2991968>.
- [65] A. Guernine, M.T. Kimour, Optimized training for convolutional neural network using enhanced Grey wolf optimization algorithm, *Informatica (Slovenia)* 45 (2021), <https://doi.org/10.11989/JEST.1674-862X.61103113>.
- [66] H. Samma, S.A. Suandi, N.A. Ismail, S. Sulaiman, L. Ping, Evolving pre-trained CNN using two-layers optimizer for road damage detection from drone images, *IEEE Access PP* (2021) 1, 11/29, <https://doi.org/10.1109/ACCESS.2021.3131231>.
- [67] Y. Yu, M. Rashidi, B. Samali, M. Mohammadi, T.N. Nguyen, X. Zhou, Crack detection of concrete structures using deep convolutional neural networks optimized by enhanced chicken swarm algorithm, *Struct. Health Monit.* 21 (5) (2022) 2244–2263, <https://doi.org/10.1177/174759217211053546>, 09/01 2022.
- [68] L. Tian, Z. Wang, W. Liu, Y. Cheng, F.E. Alsaadi, X. Liu, Empower parameterized generative adversarial networks using a novel particle swarm optimizer: algorithms and applications, *Int. J. Mach. Learn. Cybern.* 13 (4) (2022) 1145–1155, <https://doi.org/10.1007/s13042-021-01440-3>, 04/01 2022.
- [69] S.K. Baduge, et al., Assessment of crack severity of asphalt pavements using deep learning algorithms and geospatial system, *Constr. Build. Mater.* 401 (2023), <https://doi.org/10.1016/j.conbuildmat.2023.132684>, Art no. 132684.
- [70] S. Chambon, J.-M. Moliard, Automatic road pavement assessment with image processing: review and comparison, *Int. J. Geophys.* 2011 (2011), <https://doi.org/10.1155/2011/989354>.
- [71] Q. Zou, Y. Cao, Q. Li, Q. Mao, S. Wang, Crack tree: automatic crack detection from pavement images, *Pattern Recogn. Lett.* 33 (2012) 227–238, 02/01, <https://doi.org/10.1016/j.patrec.2011.11.004>.
- [72] Y. Shi, L. Cui, Z. Qi, F. Meng, Z. Chen, Automatic road crack detection using random structured forests, *IEEE Trans. Intell. Transp. Syst.* 17 (2016) 3434–3445, <https://doi.org/10.1109/TITS.2016.2552248>.
- [73] M. Eisenbach, et al., How to Get Pavement Distress Detection Ready for Deep Learning? A Systematic Approach, 2017, pp. 2039–2047, <https://doi.org/10.1109/IJCNN.2017.7966101>.
- [74] H. Maeda, Y. Sekimoto, T. Seto, T. Kashiya, H. Omata, Road damage detection and classification using deep neural networks with smartphone images, *Comput.-Aided Civil Infrastruct. Eng.* 33 (12) (2018) 1127–1141, <https://doi.org/10.1111/mice.12387>.
- [75] F. Yang, L. Zhang, S. Yu, D. Prokhorov, X. Mei, H. Ling, Feature pyramid and hierarchical boosting network for pavement crack detection, *IEEE Trans. Intell. Transp. Syst.* PP (2019) 1–11, 04/22, <https://doi.org/10.1109/TITS.2019.2910595>.
- [76] A. Lopez-Rincon, A. Tonda, M. Elati, O. Schwander, B. Piwowarski, P. Gallinari, Evolutionary optimization of convolutional neural networks for cancer miRNA biomarkers classification, *Appl. Soft Comput. J.* 65 (2018) 91–100, <https://doi.org/10.1016/j.asoc.2017.12.036>.
- [77] K. Jang, N. Kim, Y.K. An, Deep learning-based autonomous concrete crack evaluation through hybrid image scanning, *Struct. Health Monit.* 18 (5–6) (2019) 1722–1737, <https://doi.org/10.1177/1475921718821719>.
- [78] R. Augustauskas, A. Lipnickas, Improved pixel-level pavement-defect segmentation using a deep autoencoder, *Sensors (Switzerland)* 20 (9) (2020), <https://doi.org/10.3390/s20092557>.
- [79] C. Feng, H. Zhang, S. Wang, Y. Li, H. Wang, F. Yan, Structural damage detection using deep convolutional neural network and transfer learning, *KSCIE J. Civ. Eng.* 23 (10) (2019) 4493–4502, <https://doi.org/10.1007/s12205-019-0437-z>.
- [80] R.S. Rajadurai, S.T. Kang, Automated vision-based crack detection on concrete surfaces using deep learning, *Appl. Sci. (Switzerland)* 11 (11) (2021), <https://doi.org/10.3390/app11115229>.
- [81] R. Santos, D. Ribeiro, P. Lopes, R. Cabral, R. Calçada, Detection of exposed steel rebars based on deep-learning techniques and unmanned aerial vehicles, *Autom. Constr.* 139 (2022) 104324, <https://doi.org/10.1016/J.AUTCON.2022.104324>, 7// 2022.
- [82] C. Szegedy, et al., Going deeper with convolutions, in: 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 7–12 June 2015, 2015, pp. 1–9, <https://doi.org/10.1109/CVPR.2015.7298594>.
- [83] H. Xie, L. Zhang, C.P. Lim, Evolving CNN-LSTM models for time series prediction using enhanced Grey wolf optimizer, *IEEE Access* 8 (2020) 161519–161541, <https://doi.org/10.1109/ACCESS.2020.3021527>.
- [84] F. Elghaish, et al., Developing a new deep learning CNN model to detect and classify highway cracks, *J. Eng., Design Technol.* 20 (4) (2022) 993–1014, <https://doi.org/10.1108/JEDT-04-2021-0192>.
- [85] N.-D. Hoang, T.-C. Huynh, X.-L. Tran, V.-D. Tran, A novel approach for detection of pavement crack and sealed crack using image processing and Salp swarm algorithm optimized machine learning, *Adv. Civil Eng.* 2022 (2022) 9193511, <https://doi.org/10.1155/2022/9193511>, 03/15 2022.
- [86] H. Samma, S.A. Suandi, N.A. Ismail, S. Sulaiman, L.L. Ping, Evolving pre-trained CNN using two-layers optimizer for road damage detection from drone images, *IEEE Access* 9 (2021) 158215–158226, <https://doi.org/10.1109/ACCESS.2021.3131231>.
- [87] M.M. Islam, M.B. Hossain, M.N. Akhtar, M.A. Moni, K.F. Hasan, CNN based on transfer learning models using data augmentation and transformation for detection of concrete crack, *Algorithms* 15 (8) (2022), <https://doi.org/10.3390/a15080287>, Art no. 287.
- [88] P. Arafat, A. Issa, A.H.M.M. Billah, Performance Comparison of Multiple Convolutional Neural Networks for Concrete Defects Classification, *Sensors* 22 (22) (2022), <https://doi.org/10.3390/s22228714>, Art no. 8714.